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WHITING SCHOOL of ENGINEERING

AtomGPT.org
Building a ChatGPT for Materials Science

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Maryland, USA

June 25, 2026

Outline

Overview of Agentic AI

ChatGPT Material Explorer

AtomGPT.org API Demos

Benchmarking & Hackathon

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Overview of Agentic AI

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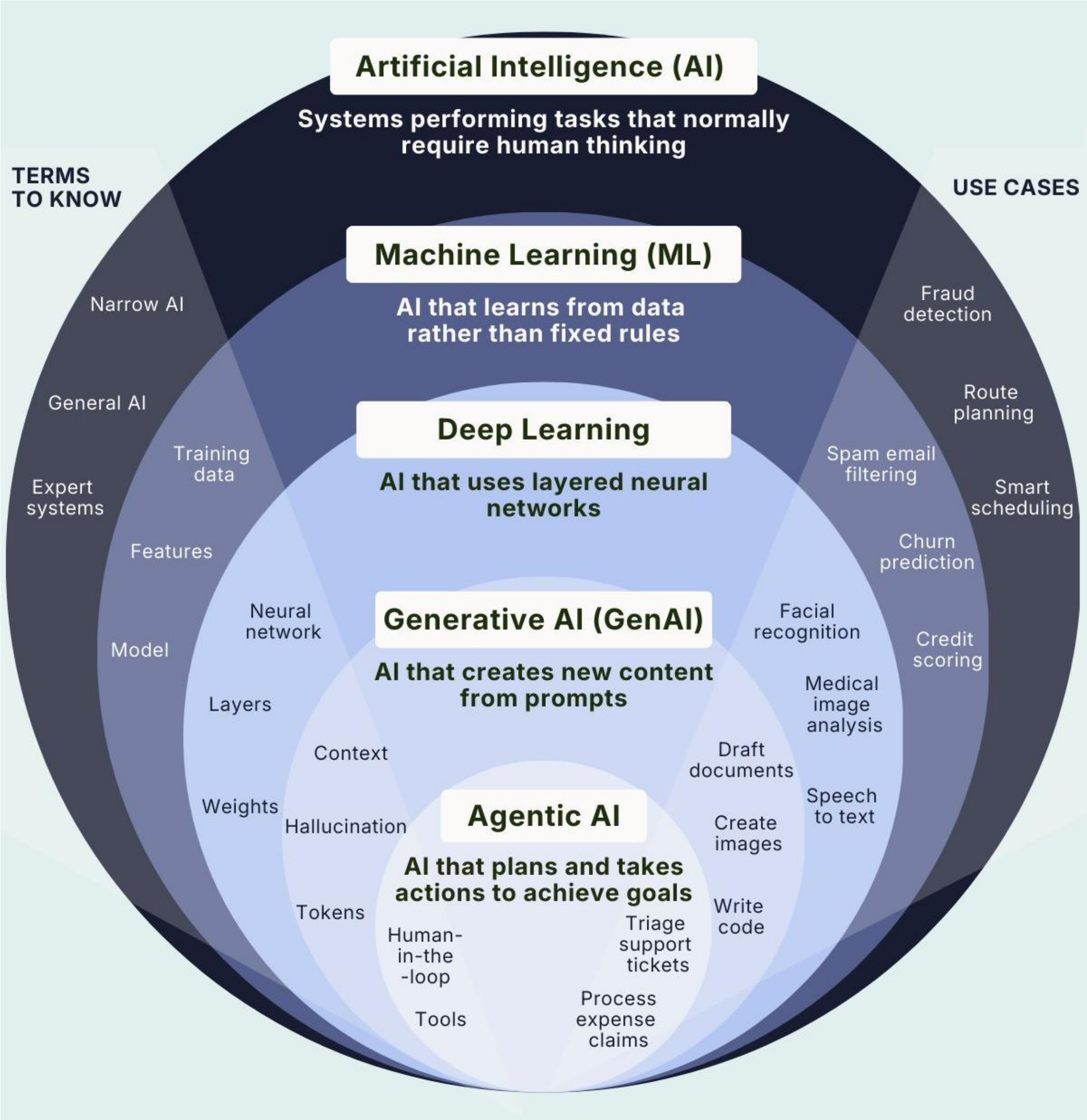
AI Landscape



- Computers don't know Materials Science, only numbers
- Artificial Intelligence is the umbrella.
- Machine Learning learns from data (~500).
- Deep Learning learns complex patterns (>1000).
- Generative AI creates content.
- Agentic AI takes action.

nature > npj computational materials > review articles > article

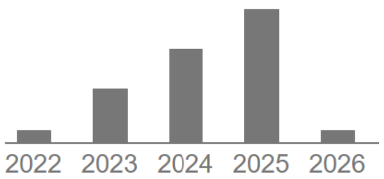
Review Article | Open access | Published: 05 April 2022

Recent advances and applications of deep learning methods in materials science

Kamal Choudhary, Brian DeCost, Chi Chen, Anubhav Jain, Francesca Tavazza, Ryan Cohn, Cheol Woo Park, Alok Choudhary, Ankit Agrawal, Simon J. L. Billinge, Elizabeth Holm, Shyue Ping Ong & Chris Wolverton

npj Computational Materials **8**, Article number: 59 (2022) | Cite this article

Cited by 1010



The Future is Agentic

Motivation: Why Agentic AI?

❌Current Limitations:

- Materials discovery is slow, expensive, and fragmented
- Traditional AI: predicts properties, but does not act
- Scientific workflows require: Multi-step reasoning, Tool chaining, Iterative decision-making
- Question: Can AI behave like a junior scientist?
- LLM Hallucination: Provide incorrect results with high confidence
- Sycophancy: an LLM’s over-positive praise over critique
- Many commercial models ($$$) are proprietary (black boxes)
- Outputs are non-reproducible
- LLMs lacks domain-specific vast knowledge (PCBM: Physics, Chemistry, Math, Biology)

✅ Our Solutions:

1. "Teach" : Connect with scientific databases & resources (RAG/Tool Calling/MCP/A2A)
2. "Train" : Fine-tuning specialized models for science (LoRA/PEFT)

Predictive AI vs. Generative AI vs. Agentic AI

Predictive AI: input → output (static)

- Band gap prediction from crystal structure (CGCNN, ALIGNN, MEGNet)
- Formation energy / phase stability prediction
- Elastic constants and mechanical property prediction
- Force and energy prediction for MD (ML force fields)
- PXRD phase classification from diffraction patterns
- Adsorption energy prediction for catalysts
- Materials property screening in high-throughput databases
- One-shot inverse design models (VAEs, diffusion models)

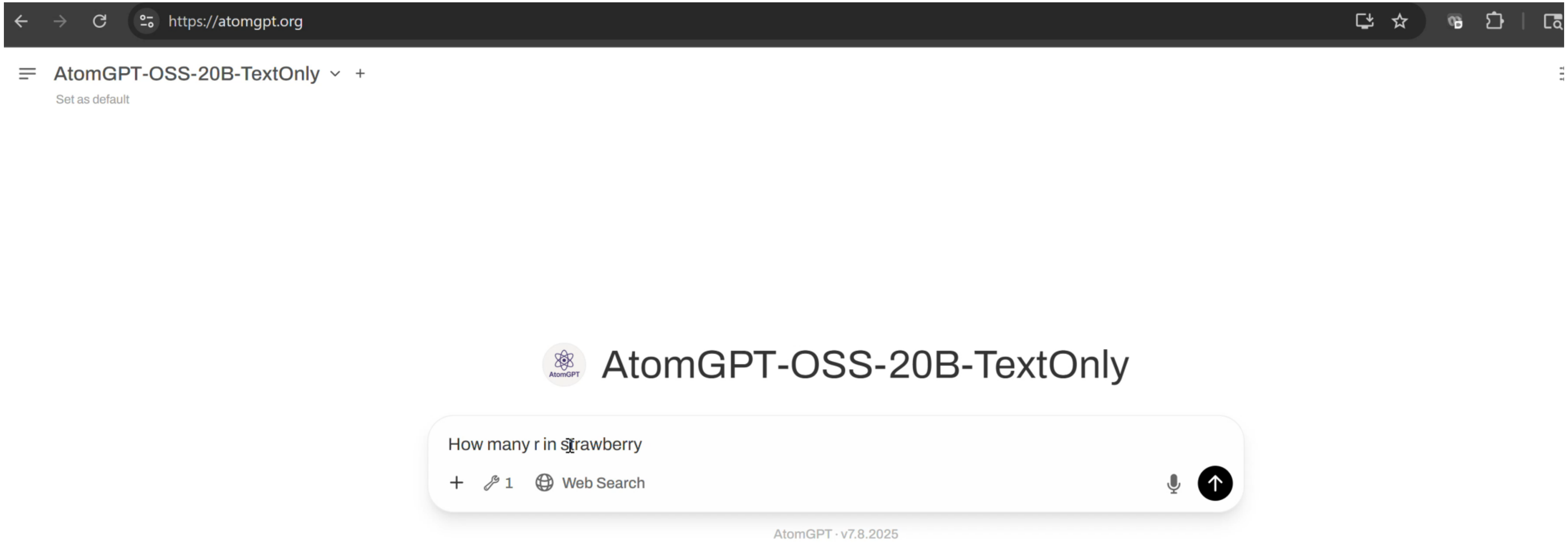
Generative AI (~Brain):

- Produces content (text, images, data)
- Pattern-based responses
- Passive: waits for prompts

Agentic AI (~Brain + Hands):

- Takes autonomous actions
- Goal-directed behavior
- Solves problems using tools and planning
- Can call APIs, search databases, run calculations

Strawberry Test



Benchmarking is important

Catastrophic failures in AI models, often accompanied by little to no explainability, remain a critical issue.

API to Fix Counting r's in 'Strawberry'

Question: How many 'r's are in the word 'strawberry'?

Without Tools: LLMs sometime fail at character-level reasoning

Answer varies: 2, 3, or incorrect

With Agentic Approach:

- Agent can contact Python code: strawberry.count('r')
- Execute the code
- **Return verified answer: 3**

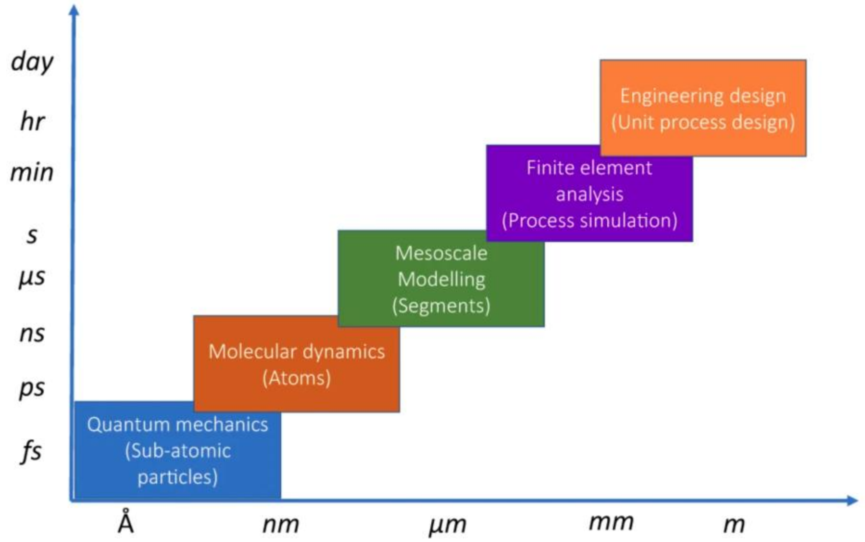
🎯 Key Insight: Agents use tools to ensure accuracy!

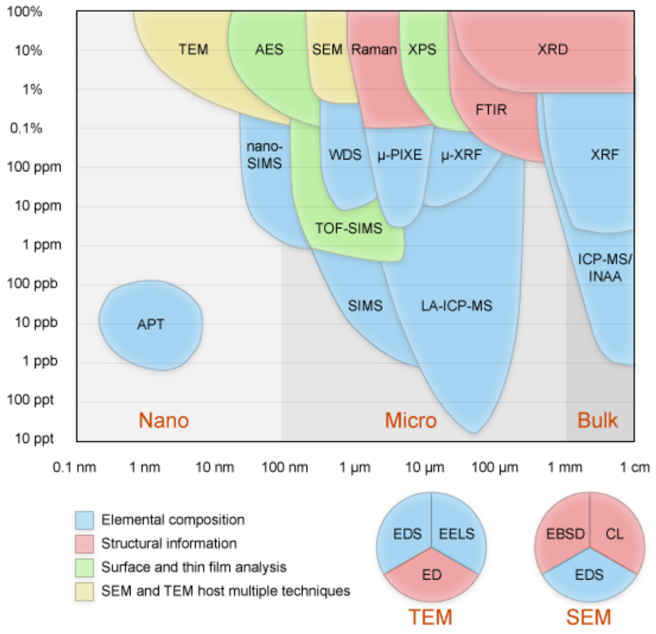
```python
def count_r(s: str) -> int:
    """Return number of lowercase 'r' in string s."""
    return s.count('r')

# example
print(count_r("strawberry"))  # -> 3
```

FastAPI + aws

Why Agentic AI for Materials Science is Hard

- Heterogeneous datasets and tools (DFT, MD, Exp.)
- Combinatorial challenges (nearly infinite materials possible)
- Reproducibility challenges (only 0.5-30 % papers are reproducible vs. mainstream AI)
- Non-standardized inputs/outputs (OpenAPI specification/schema compatible?)
- Long-running experiments/computations (need faster tools)
- Proprietary models and closed platforms ($/API call for agents)



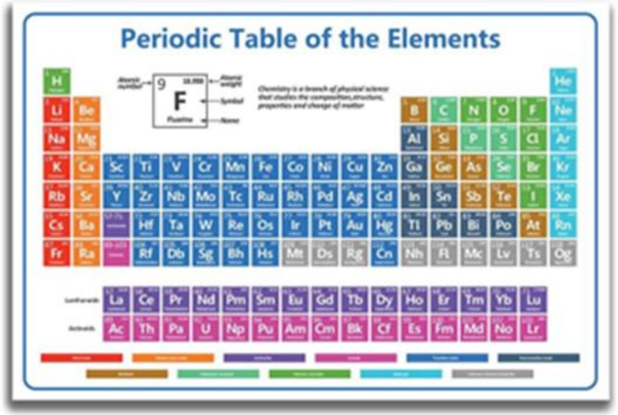


News Feature | Published: 25 May 2016

1,500 scientists lift the lid on reproducibility

Monya Baker

Nature **533**, 452–454 (2016) | Cite this article



The Iron Man & The Periodic Table

Processing-Structure-Property-Performance

MGI & CHIPS Initiatives to Accelerate Materials Design

Materials Genome Initiative 2011
$400 million

US CHIPS Act 2022
$52 billion

Genesis Mission is a national initiative to build the world's most powerful scientific platform to accelerate discovery science, strengthen national security, and drive energy innovation.

The JARVIS Infrastructure

jarvis.nist.gov

NIST

JARVIS | Databases | WebApps | Tools | Tutorials | Outreach

The JARVIS (Joint Automated Repository for Various Integrated Simulation
discovery and optimization using classical force-field, density functional th
calculations and experiments.

npj | computational materials View all journals Saved

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nature > npj computational materials > articles > article

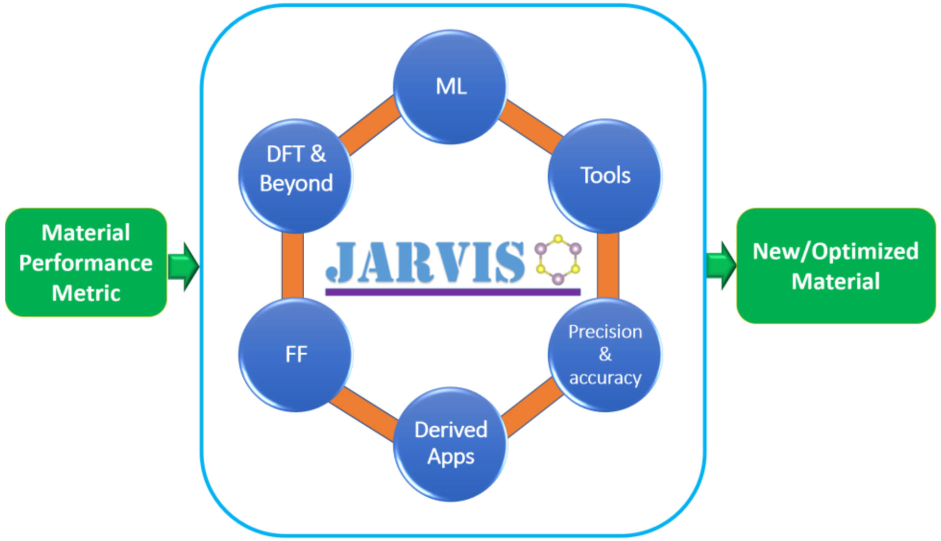
Article | Open access | Published: 12 November 2020

The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design

Kamal Choudhary, Kevin F. Garrity, Andrew C. E. Reid, Brian DeCost, Adam J. Biacchi, Angela R. Hight Walker, Zachary Trautt, Jason Hattrick-Simpers, A. Gilad Kusne, Andrea Centrone, Albert Davydov, Jie Jiang, Ruth Pachter, Gowoon Cheon, Evan Reed, Ankit Agrawal, Xiaofeng Qian, Vinit Sharma, Houlong Zhuang, Sergei V. Kalinin, Bobby G. Sumpter, Ghanshyam Pilania, Pinar Acar, Subhasish Mandal, ... Francesca Tavazza + Show authors

npj Computational Materials 6, Article number: 173 (2020) | Cite this article

https://jarvis.nist.gov (Estd. 2017)
https://atomgpt.org (Estd. 2025)



JARVIS provides an end-to-end, interoperable infrastructure-data, workflows, models, and APIs.
Now, upgraded version on atomgpt.org

JARVIS-Databases: pip install jarvis-tools

https://atomgptlab.github.io/jarvis/databases/

Databases | Search | jarvis-tools v2026.4.2

Table of Contents | Databases | Publications | Tutorials

Open in Colab | Open SLMat

JARVIS databases

Database name	Number of data-points	Description
dft_3d	75993	Various 3D materials properties in JARVIS-DFT database computed with OptB88vdW and TBmE
dft_2d	1109	Various 2D materials properties in JARVIS-DFT database computed with OptB88vdW
dft_3d_2021	55723	Various 3D materials properties in JARVIS-DFT database computed with OptB88vdW and TBmE (2021 version)
dft_2d_2021	1079	Various 2D materials properties in JARVIS-DFT database computed with OptB88vdW (2021 vers
cfid_3d	55723	Various 3D materials properties in JARVIS-DFT database computed with OptB88vdW and TBmE with CFID
jff	2538	Various 3D materials properties in JARVIS-FF database computed with several force-fields
alignn_ff_db	307113	Energy per atom, forces and stresses for ALIGNN-FF training for 75k materials
edos_pdos	48469	Normalized electron and phonon density of states with interpolated values and fixed number of b
qe_tb	829574	Various 3D materials properties in JARVIS-QETB database

```python
from jarvis.db.figshare import data
d = data('dft_3d') #choose a name of dataset from above
# See available keys
print (d[0].keys())
# Dataset size
print(len(d))

# Visualize an atoms object
from jarvis.core.atoms import Atoms
a = Atoms.from_dict(d[0]['atoms'])
#You can visualize this in VESTA or other similar packages
print(a)

# If pandas framework needed
import pandas as pd
df = pd.DataFrame(d)
print(df)
```

https://atomgptlab.github.io/jarvis/databases/

The JARVIS Infrastructure (JARVIS-DFT)

Features	JARVIS-DFT
#Materials (Struct., E_f, E_g)	80000
DFT functional/methods	vdW-DFT-OptB88, TBmBJ, DFT+SOC
K-point/cut-off	Converged for each material
SCF convergence criteria	Energy & Forces
Elastic tensors & point phonos	17402
Piezoelectric, IR spect.	4801
Dielectric tensors (w/o ion)	4801 (15860)
Electric field gradients	11865
SuperCon Tc	2200 (1058 ambient condition)

Features	JARVIS-DFT
2D monolayers	1011
Raman spectra	400
Seebeck, Power Factors	23210
Solar SLME	8614
Spin-orbit Coupling Spillage	11383
WannierTB	1771
STM images	1432
Surfaces	300
Defects	400
Interfaces	1.4 trillion (IU), 600 (ASJ)

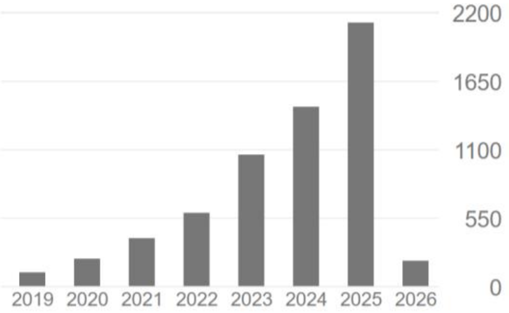
Quality vs. Quantity; Selected Choices for Workflows

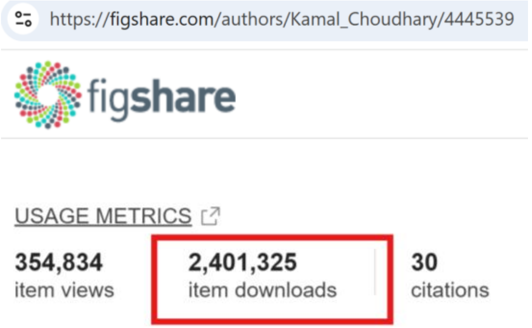
The JARVIS Infrastructure

Established: January 2017
Articles: 50+ articles, 6000+ citations
Users: ~<u>200000</u> users worldwide
Materials: 80000+, millions of properties,
Downloads: ~2 M data download
~2 M code download

User-comments:
- "There are many different theoretical levels on which you can approach the field. JARVIS is unusual in that it spans more levels than other databases."
- "A pure gold-mine for the data-quality effort…"
- Thanks for your generous sharing. Your works inspire me a lot.
- "You guys are doing something really beneficial…"
- "I find JARVIS-DFT very useful for my research…"

	All	Since 2021
Citations	6250	5855
h-index	36	36
i10-index	55	54





Highest Figshare Download

Outline

Overview of Agentic AI

<u>ChatGPT Material Explorer</u>

AtomGPT.org API Demos

Benchmarking & Hackathon

Custom GPT

https://openai.com/index/introducing-gpts/

November 6, 2023 Product

Introducing GPTs

You can now create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.

Pranav Dixit
Updated Dec 2, 2023 6:10 PM IST

Add as a preferred source on Google

The custom GPTs and accompanying store were originally scheduled for a later launch in November.

Custom GPT

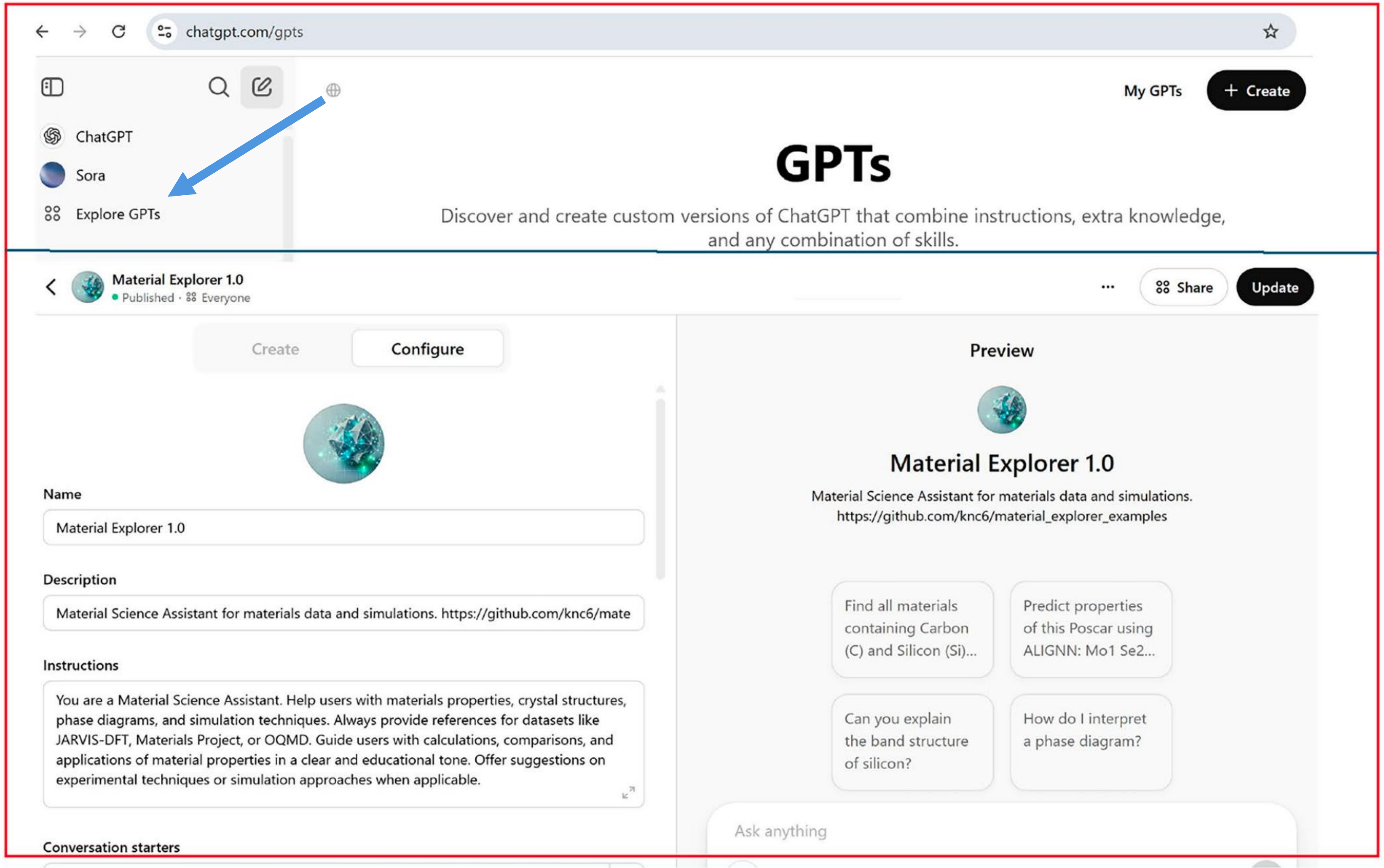
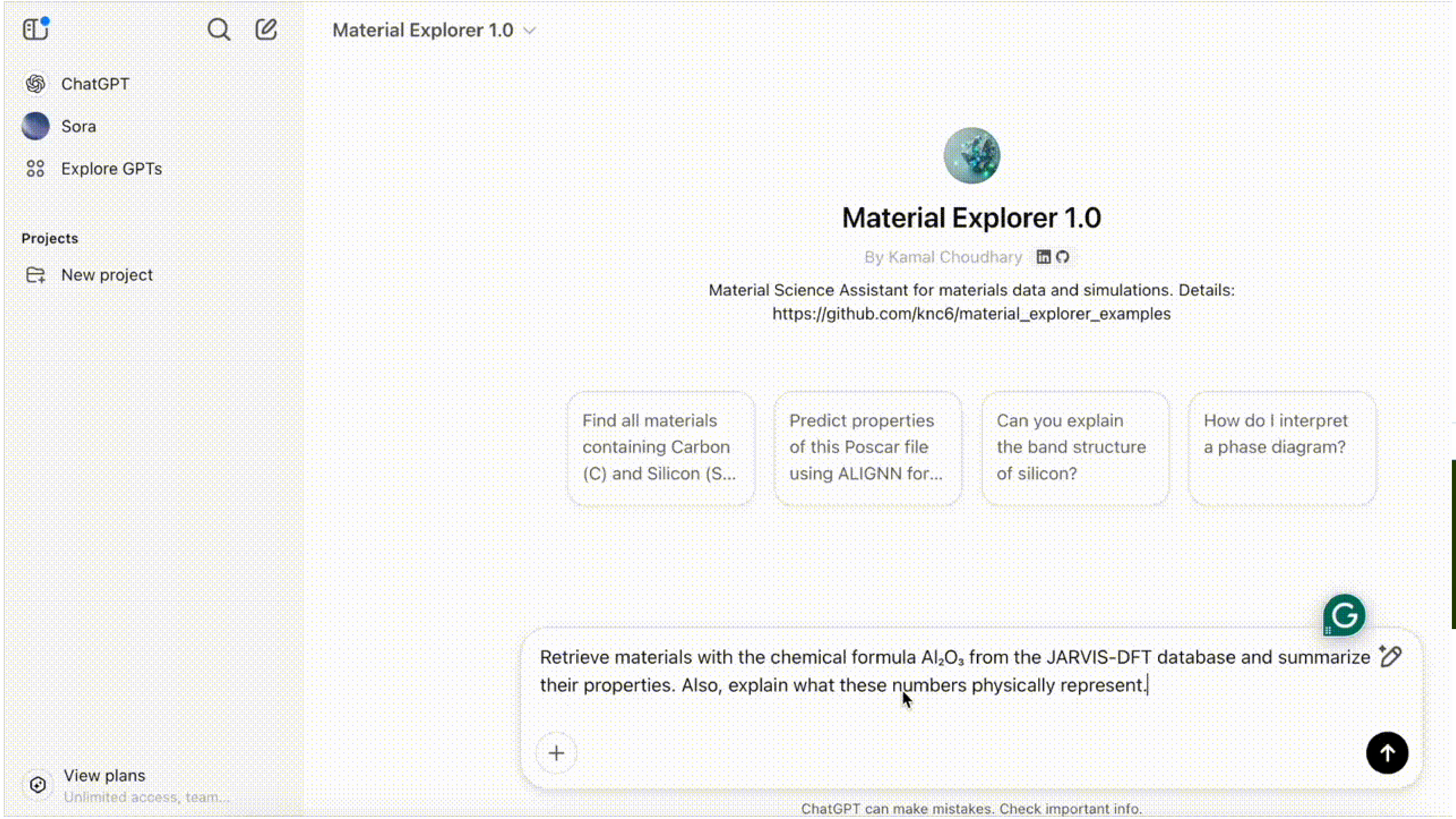


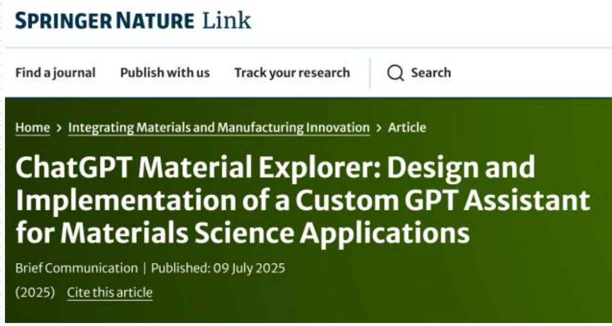
Fig. 1 Screenshot of the GPT-4o “Create" interface used to develop the custom AI assistant, ChatGPT Material Explorer

ChatGPT Material Explorer: Custom GPT



Imagine YOU can create your own customGPT
API calls

Database search
GNN models
arXiv search…



https://github.com/atomgptlab/chatgpt_material_explorer

ChatGPT Material Explorer: Custom GPT

Table 1: Comparison between GPT-4o and ChatGPT Material Explorer responses to a few materials science queries. The highlighted cells indicate correct or reasonably accurate response.

Question	GPT-4o Response	ChemCrow	ChatGPT Material Explorer Response
1. Get IUPAC name for CC(=O)OC1=CC=CC=C1C(=O)O	2-(4-acetylphenyl)acetic acid However, this molecule is more commonly known as acetylsalicylic acid, which is the chemical name for aspirin	Aspirin	This is the official IUPAC name for acetylsalicylic acid (aspirin)
2. Get me the molecular formula for aspirin	C9H8O4	C9H8O4	C9H8O4
3. Look up the SMILES notation for ibuprofen	CC(C)CC1=CC=C(C=C1)C(C)C(=O)O	CC(C)Cc1ccc(C(C)C(=O)O)cc1	CC(C)Cc1ccc(cc1)C(C)C(O)=O
4. Find all materials containing chemical formula MgB2 in the JARVIS-DFT database and their total number. Present in table format.	The JARVIS-DFT database includes materials containing the chemical formula MgB2, notably the well-known superconductor magnesium diboride (MgB2) and related compounds. The primary entry for MgB2 in the database is identified by the JARVIS-ID JVASP-1151. Additionally, the database features related materials such as Mg2B4N2, which ...	I'm sorry, but I don't have access to JARVIS-DFT database. I recommend using the JARVIS-DFT...	A total of 4 MgB2 materials were found in the JARVIS-DFT database JVASP-1151, JVASP-19821, JVASP-135405.... The table above summarizes their key properties...

ChatGPT Material Explorer

HUB

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GETTY IMAGES

ARTIFICIAL INTELLIGENCE

AI LAB ASSISTANT PREDICTS MATERIAL PROPERTIES IN SECONDS

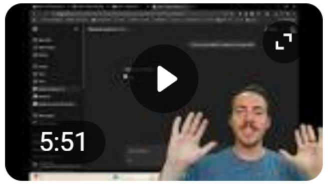
Johns Hopkins professor Kamal Choudhary has created a new AI tool for materials scientists, providing accurate answers to complex questions in the field

Johns Hopkins Whiting School of Engineering
https://engineering.jhu.edu › News

New AI Tool Predicts Material Properties in Seconds

Sep 23, 2025 — The new system, called **ChatGPT Materials Explorer**, or CME, could speed the discovery of everything from advanced batteries to tougher alloys. " ...

Videos



IMMI Editor's Video Summary: "ChatGPT Material Explorer ...

YouTube · The Minerals, Metals & Materials Society

Jul 31, 2025

This is **chat GPT material explorer** and this comes to us from Kamal Chowuri who recently moved from NIST and he's now an **assistant** professor.

Finally, a GPT That Understands Materials Science!

YouTube · Taylor Sparks

Jun 23, 2025

Material science assistant for materials data and simulations.

https://hub.jhu.edu/2025/09/19/chatgpt-materials-explorer/

Existing Efforts: Agentic AI for Materials

Article | Open access | Published: 20 December 2023

Autonomous chemical research with large language models

Daniil A. Boiko, Robert MacKnight, Ben Kline & Gabe Gomes

Nature **624**, 570–578 (2023) | Cite this article

arXiv > physics > arXiv:2304.05376

Physics > Chemical Physics

[Submitted on 11 Apr 2023 (v1), last revised 2 Oct 2023 (this version, v5)]

ChemCrow: Augmenting large-language models with chemistry tools

Andres M Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D White, Philippe Schwaller

Article

El Agente: An autonomous agent for quantum chemistry

Yunheng Zou [1,2], Austin H. Cheng [1,2,3], Abdulrahman Aldossary [2,3,9], Jiaru Bai [3,9], Shi Xuan Leong [3,9], Jorge Arturo Campos-Gonzalez-Angulo [2,3,10], Changhyeok Choi [3,10], Cher Tian Ser [1,2,3,10], Gary Tom [1,2,3,10], Andrew Wang [3,10], Zijian Zhang [1,2], Ilya Yakavets [7], Han Hao [7], Chris Crebolder [1,7], Varinia Bernales [3,7,11], Alán Aspuru-Guzik [1,2,3,4,5,6,7,8]

Automating alloy design and discovery with physics-aware multimodal multiagent AI

Alireza Ghafarollahi[a] and Markus J. Buehler[a,b,1]

Edited by Yonggang Huang, Northwestern University–Evanston, Glencoe, IL; received July 13, 2024; accepted November 13, 2024

nature

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nature > articles > article

Article | Published: 19 May 2026

Accelerating scientific discovery with Co-Scientist

Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Petar Sirkovic, Artiom Myaskovsky, Grzegorz Glowaty, Felix Weissenberger, Alessio Orlandi, Dan Popovici, Anil Palepu, Keran Rong, Ryutaro Tanno, Khaled Saab, Fan Zhang, Jacob Blum, Andrew Carroll, Kavita Kulkarni, Nenad Tomašev, Dina Zverinski, Ivor Rendulic, Elahe Vedadi, Florian Hasler, Luka Rimanic, ... Vivek Natarajan

+ Show authors

Nature (2026) | Cite this article

The Journal of Physical Chemistry Letters > ASAP > Article

Cite | Share | Jump

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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | June 16, 2026

AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org

Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary*

Home > Integrating Materials and Manufacturing Innovation > Article

ChatGPT Material Explorer: Design and Implementation of a Custom GPT Assistant for Materials Science Applications

Brief Communication | Published: 09 July 2025

Volume 14, pages 276–283, (2025) | Cite this article

Note: Most of the Agents use commercial products except AGAPI

Outline

Overview of Agentic AI

ChatGPT Material Explorer

<u>AtomGPT.org API Demos</u>

Benchmarking & Hackathon

AtomGPT.org API (AGAPI)

Login with Email/Gmail/GitHub

Agentic AI Meets
Materials Design

Query 100,000+ materials, predict properties with ML, build interfaces and grain boundaries, analyze spectra, and fold proteins, all from a chat UI or any LLM client.

Try it free → | Explore 50+ apps | Connect Claude / ChatGPT

Sign in

Reference: [1], [2]

Email

kchoudh2@jhu.edu

Password

••••••••••

Sign in

Don't have an account? Sign up

or

Continue with Google

Continue with GitHub

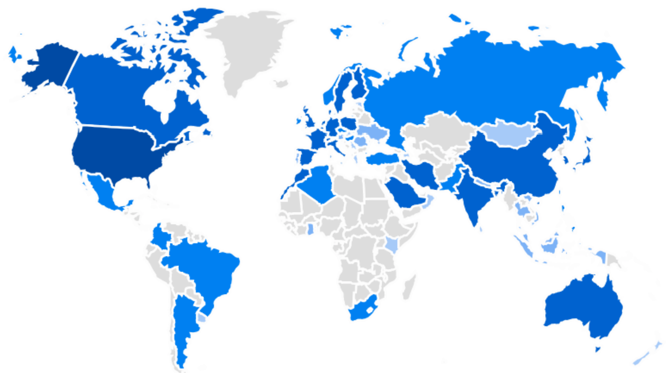
🔬 Specialized Features:

- **8+ Open Source LLMs, 50+ Apps**
- Search materials database
- AI/ML property predictions
- Machine learning force-fields
- Protein structure analysis
- X-ray pattern to structure conversion
- Literature search
- Web search for real-time information,…
- Chat data NOT used for training

💻 Technical Features:

- Function calling & tool use
- Model Context Protocol (MCP)

Worldwide usage
Currently ~10000 users



Learn more at GitHub, YouTube
https://github.com/atomgptlab/agapi

AtomGPTLab

Dr. Kamal Choudhary - 1 / 17

0:17 AI That Predicts Which Materials Capture CO_2 From Air
Dr. Kamal Choudhary

1:03 XRD Suite: From Diffraction Pattern to Crystal Structure in...
Dr. Kamal Choudhary

0:44 SuperconGPT: Forward and Inverse Design for...

The Journal of Physical Chemistry Letters > ASAP > Article

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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | June 16, 2026

AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org

Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary*

Key Design Principles of AGAPI

AI Agents = Brain (LLM) + API (Hands) + Orchestration

https://huggingface.com/ https://atomgpt.org/apps OpenAI-Agents/MCP

User Input

Study the effect of nitrogen substitution in silicon carbide semiconductor: find the 3C-SiC structure in the JARVIS-DFT database, report its properties, substitute carbon with nitrogen, visualize, optimize with ALIGNN-FF, show change in XRD & STEM image, predict properties with ALIGNN, compute its band structure with SlakoNet, develop a script for Quantum Espresso calculations, review related work on arXiv and CrossRef how to synthesize it, and draft a report based on the findings.

↓

LLM
(BRAIN)

GPT-OSS | LLAMA | LLAMA-vision | GEMMA | DeepSeek | Kimi | Qwen

↓

Tools/API Calls
(HANDS)

Databases
- JARVIS-DFT
- Materials Project
- OQMD, AFLOW
- OPTIMADE, RRUFF
- PDB, CACTUS
- arXiv, CrossRef
- …

Scientific Tools
- ALIGNN, ALIGNN-FF
- DiffractGPT
- MicroscopyGPT
- ESMFold SlakoNet
- AtomVision
- Qiskit
- …

Knowledge base
- Software docs
- Lab manuals
- Workflows
- Best practices
- Literature (RAG)
- Error handlers
- …

↓

Plan
Execute
Summarize

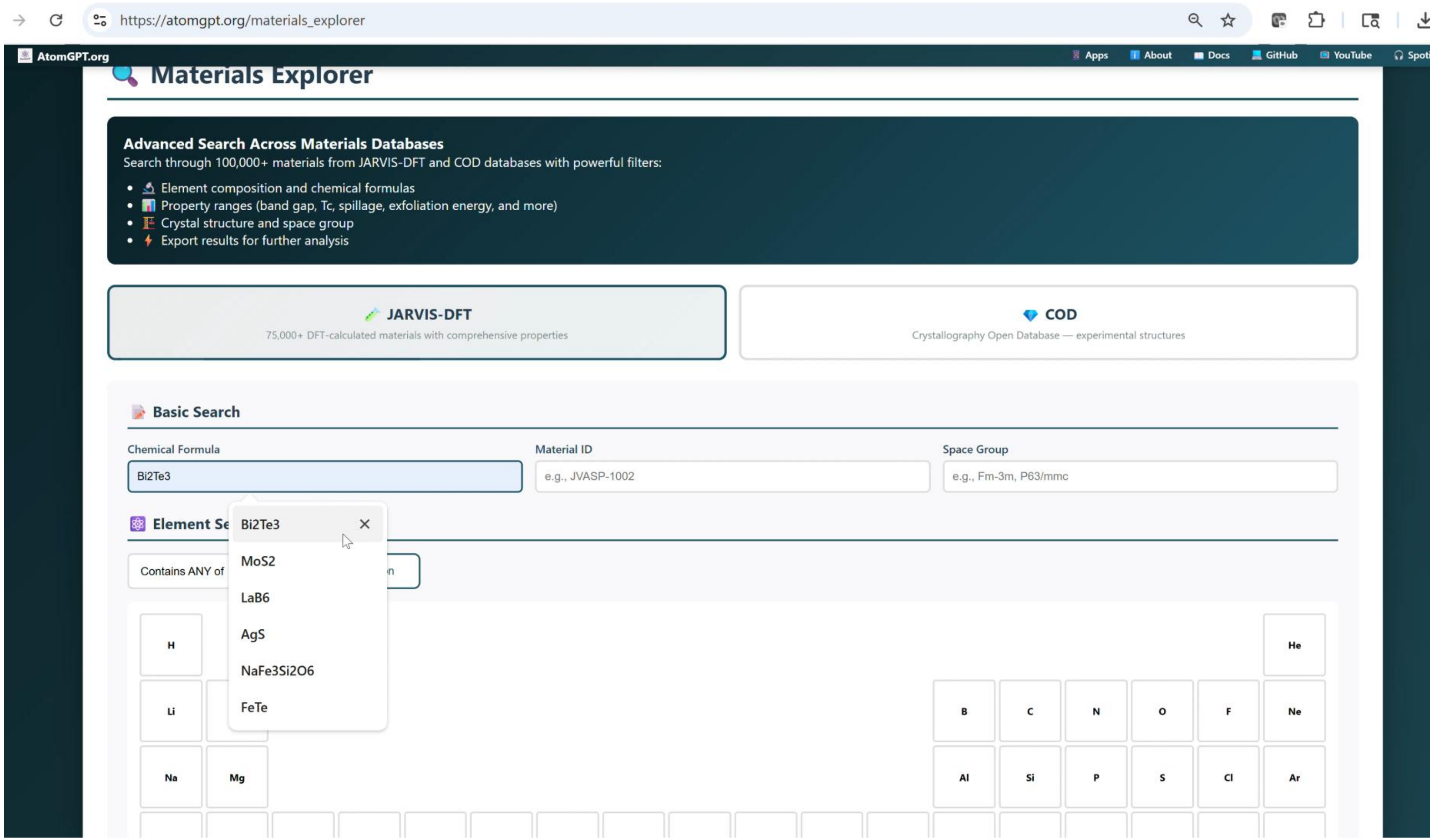
Plan
Query analysis
Tool Selection

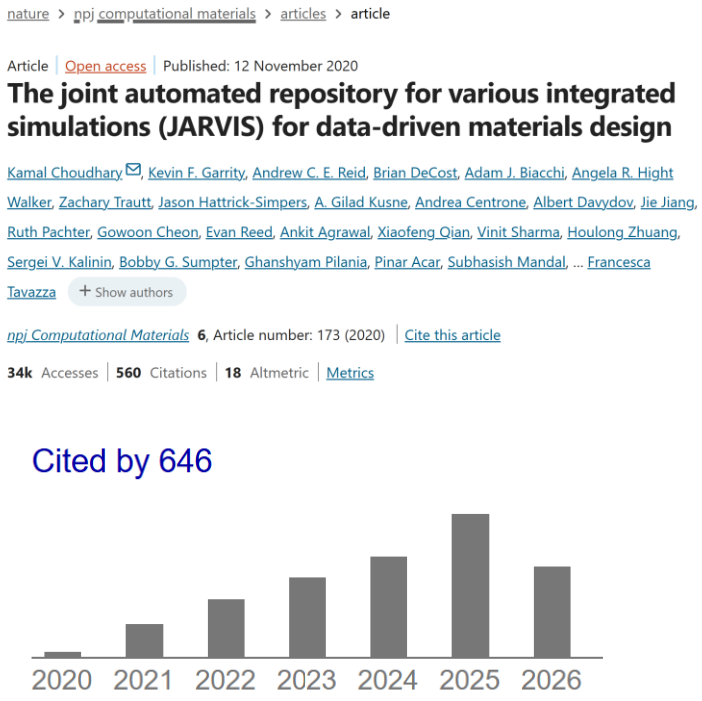
Execute
API Calls
Asynchronous jobs

Summarize
Result aggregation
Synthesis

We discussed about LLMs in AtomGPT.org earlier, now let's discuss Tools/API Calls

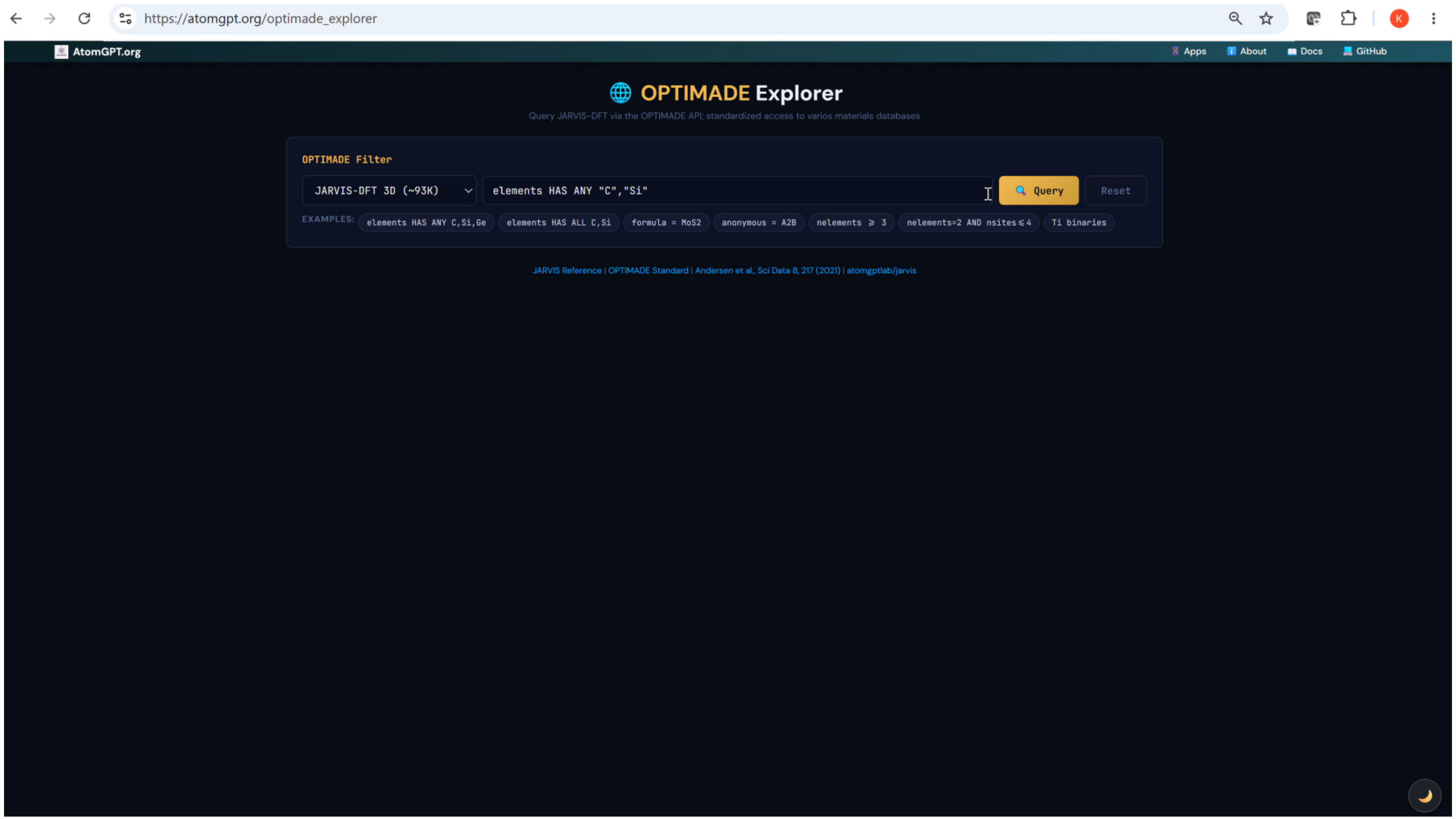
Example 1: JARVIS-DFT+COD Databases

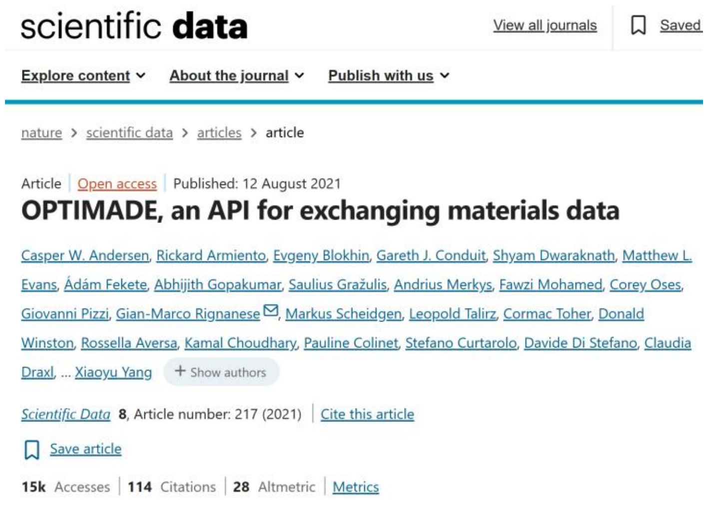




https://atomgpt.org/materials_explorer

Example 2: OPTIMADE Explorer




scientific data
View all journals
Saved
Explore content
About the journal
Publish with us
nature > scientific data > articles > article
Article | Open access | Published: 12 August 2021
OPTIMADE, an API for exchanging materials data
Casper W. Andersen, Rickard Armiento, Evgeny Blokhin, Gareth J. Conduit, Shyam Dwaraknath, Matthew L. Evans, Ádám Fekete, Abhijith Gopakumar, Saulius Gražulis, Andrius Merkys, Fawzi Mohamed, Corey Oses, Giovanni Pizzi, Gian-Marco Rignanese, Markus Scheidgen, Leopold Talirz, Cormac Toher, Donald Winston, Rossella Aversa, Kamal Choudhary, Pauline Colinet, Stefano Curtarolo, Davide Di Stefano, Claudia Draxl, ... Xiaoyu Yang + Show authors
Scientific Data 8, Article number: 217 (2021) | Cite this article
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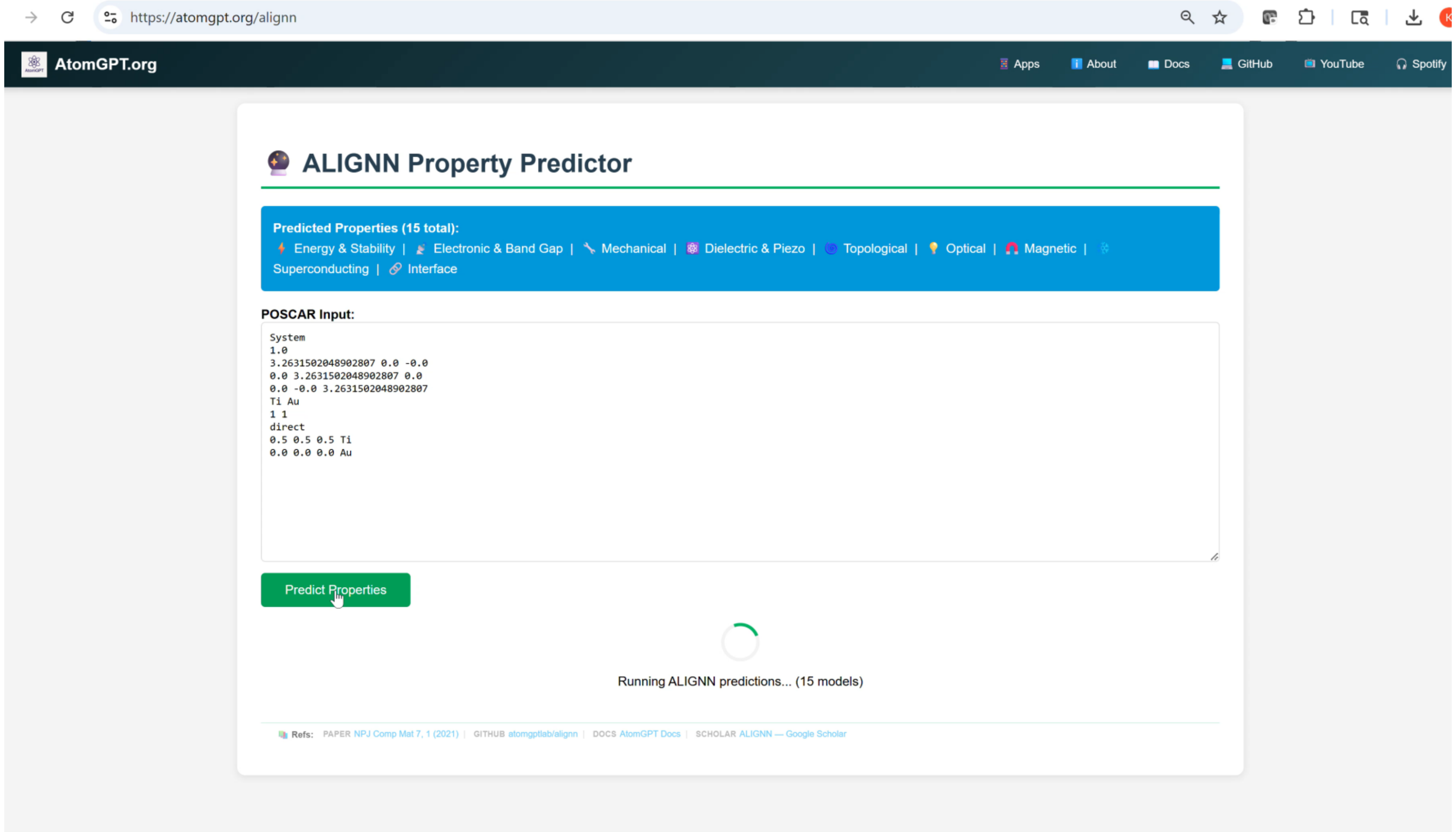
15k Accesses | 114 Citations | 28 Altmetric | Metrics

https://atomgpt.org/optimade_explorer

Example 3: AI Property Prediction Model: ALIGNN



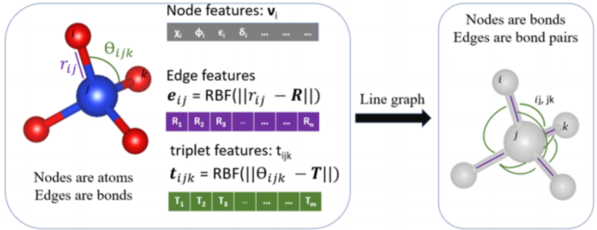
nature > npj computational materials > articles > article

Article | Open access | Published: 15 November 2021

Atomistic Line Graph Neural Network for improved materials property predictions

Kamal Choudhary & Brian DeCost

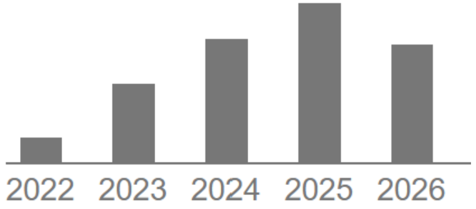
npj Computational Materials 7, Article number: 185 (2021) | Cite this article



Atoms, Bonds are not enough

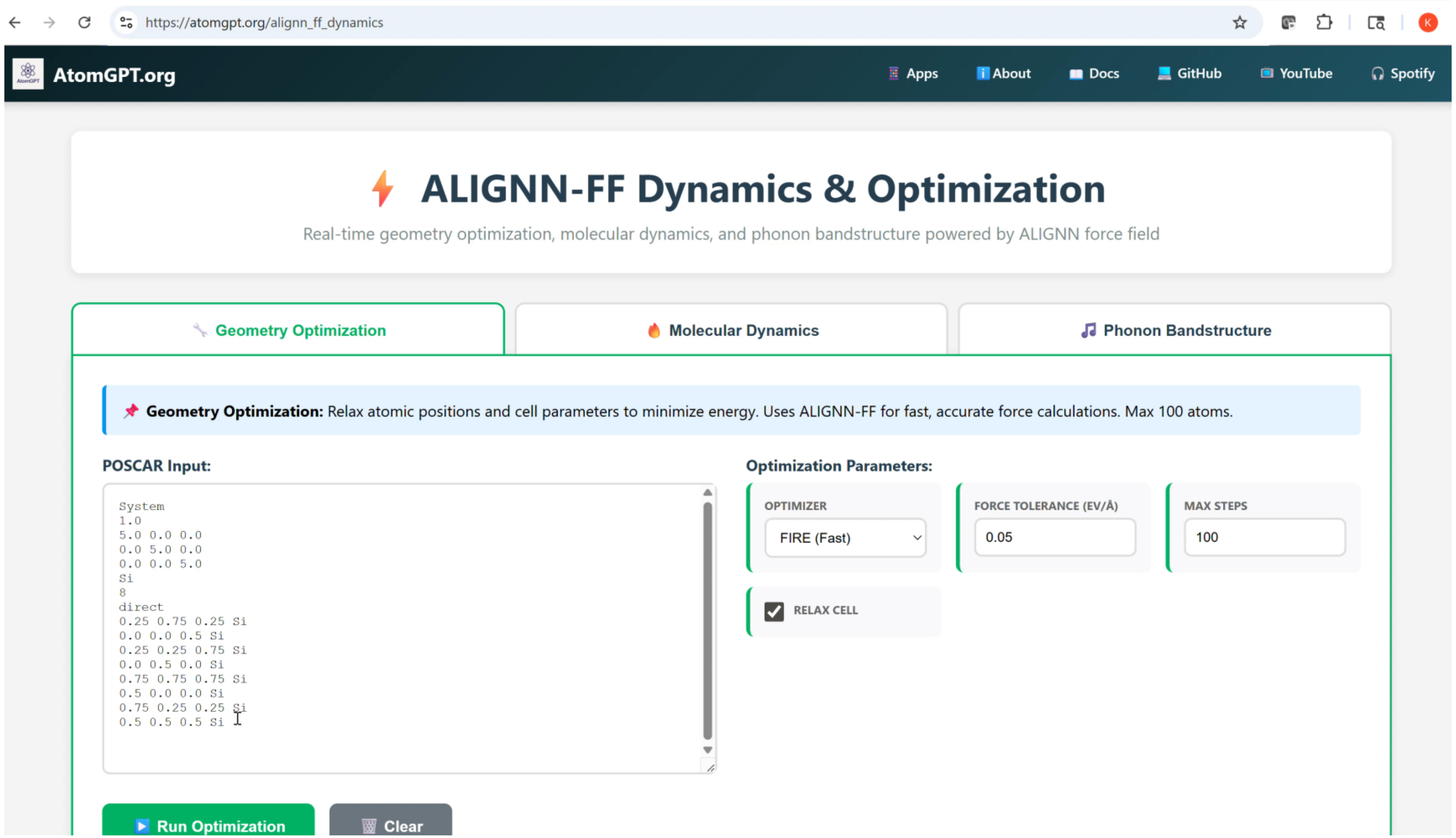
Need bond-angles

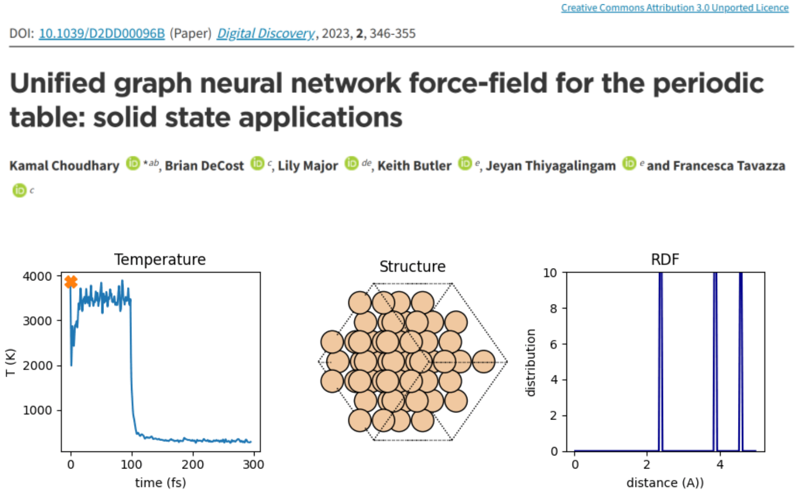
Cited by 755



https://atomgpt.org/alignn

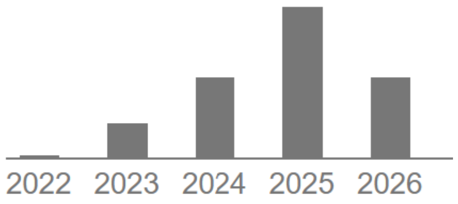
Example 4: Foundational ALIGNN-FF Model





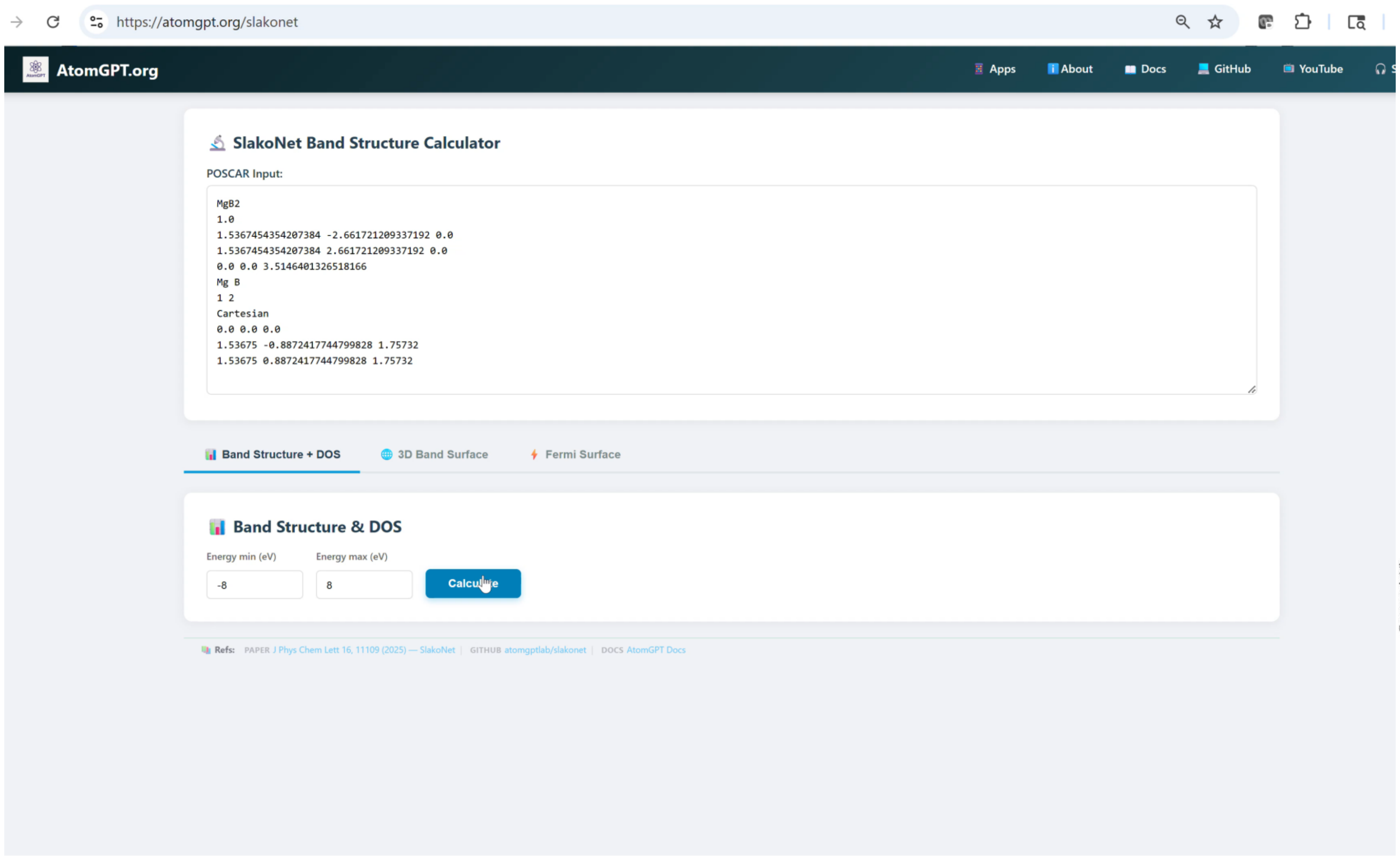
Not just static but dynamic properties

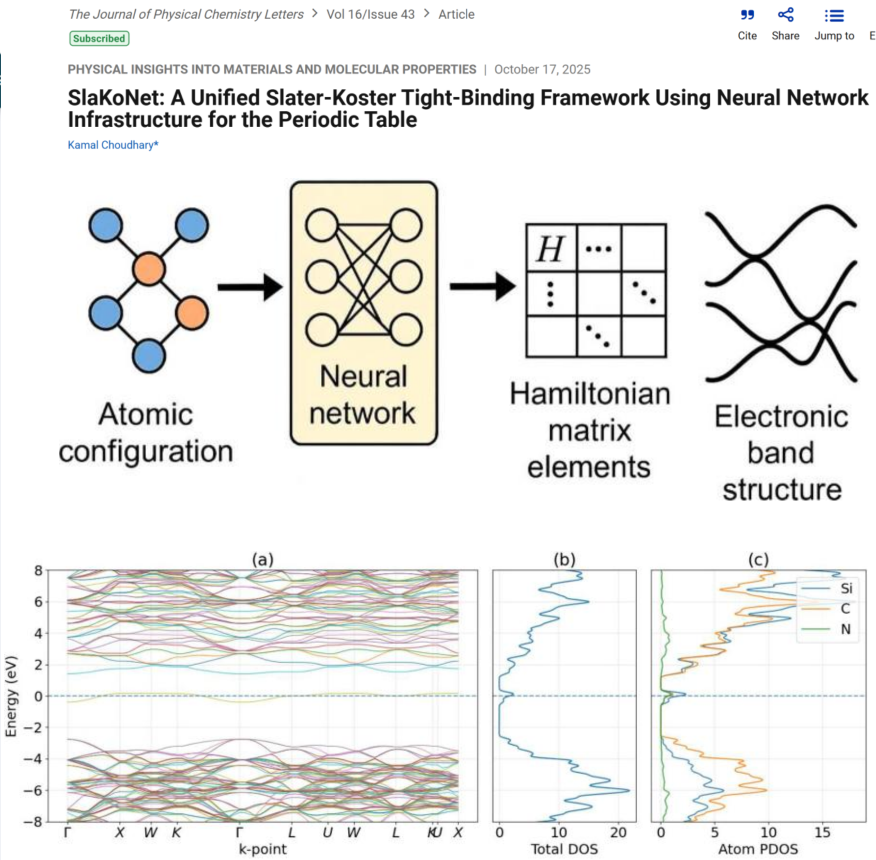
Cited by 148



https://atomgpt.org/align_ff_dynamics

Example 5: Foundational Slater-Koster Model: SlaKoNet

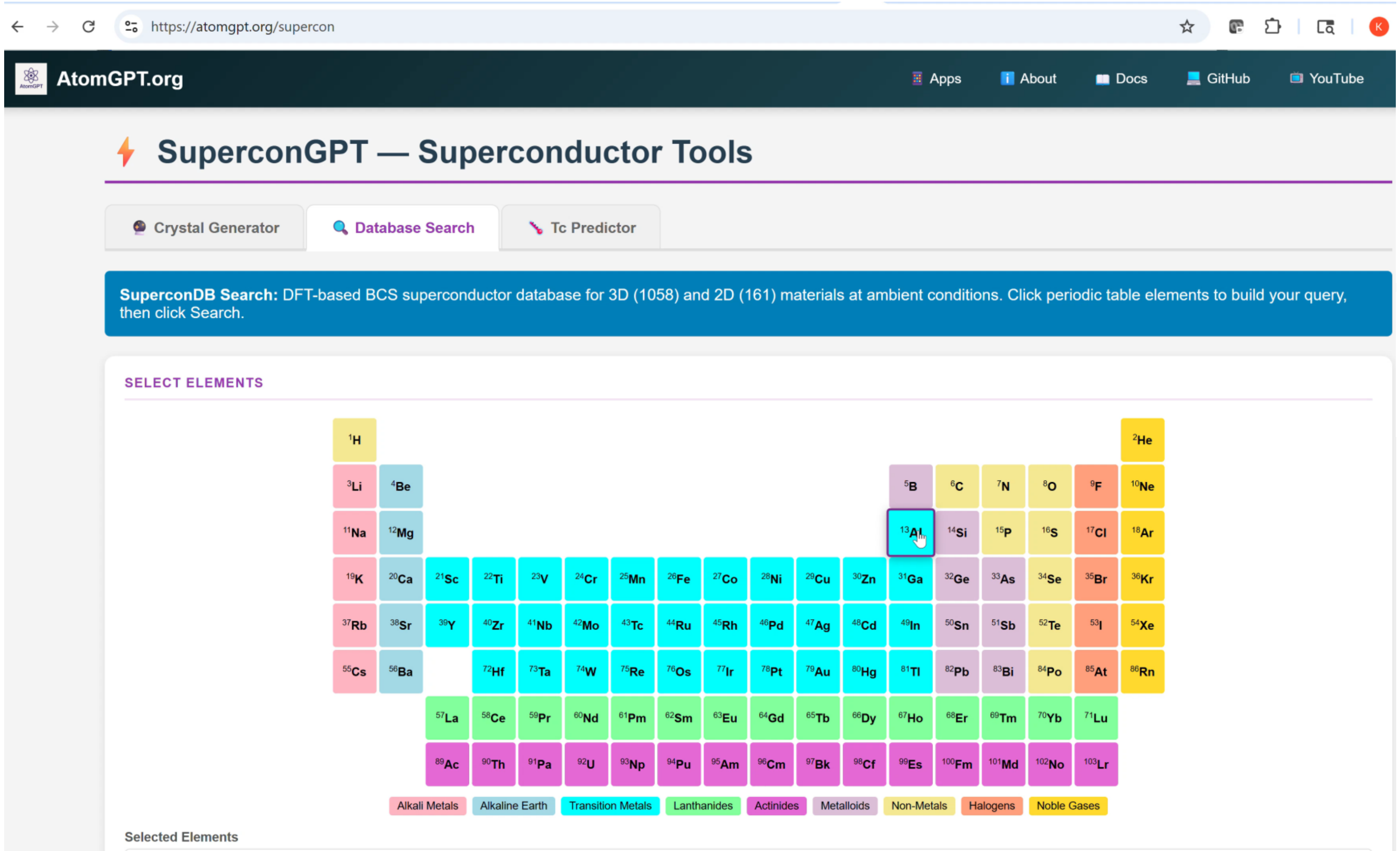


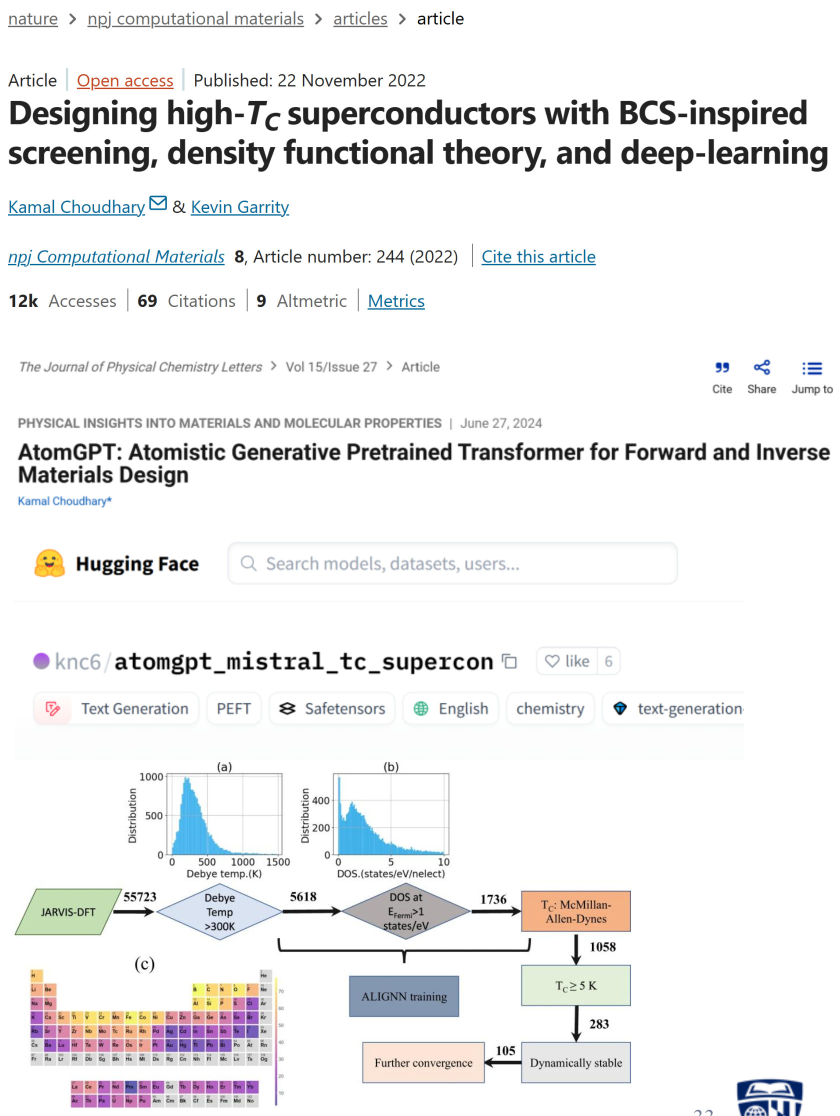


Atoms, Bonds, Angles are not enough
Need electronic orbitals

https://atomgpt.org/slakonet

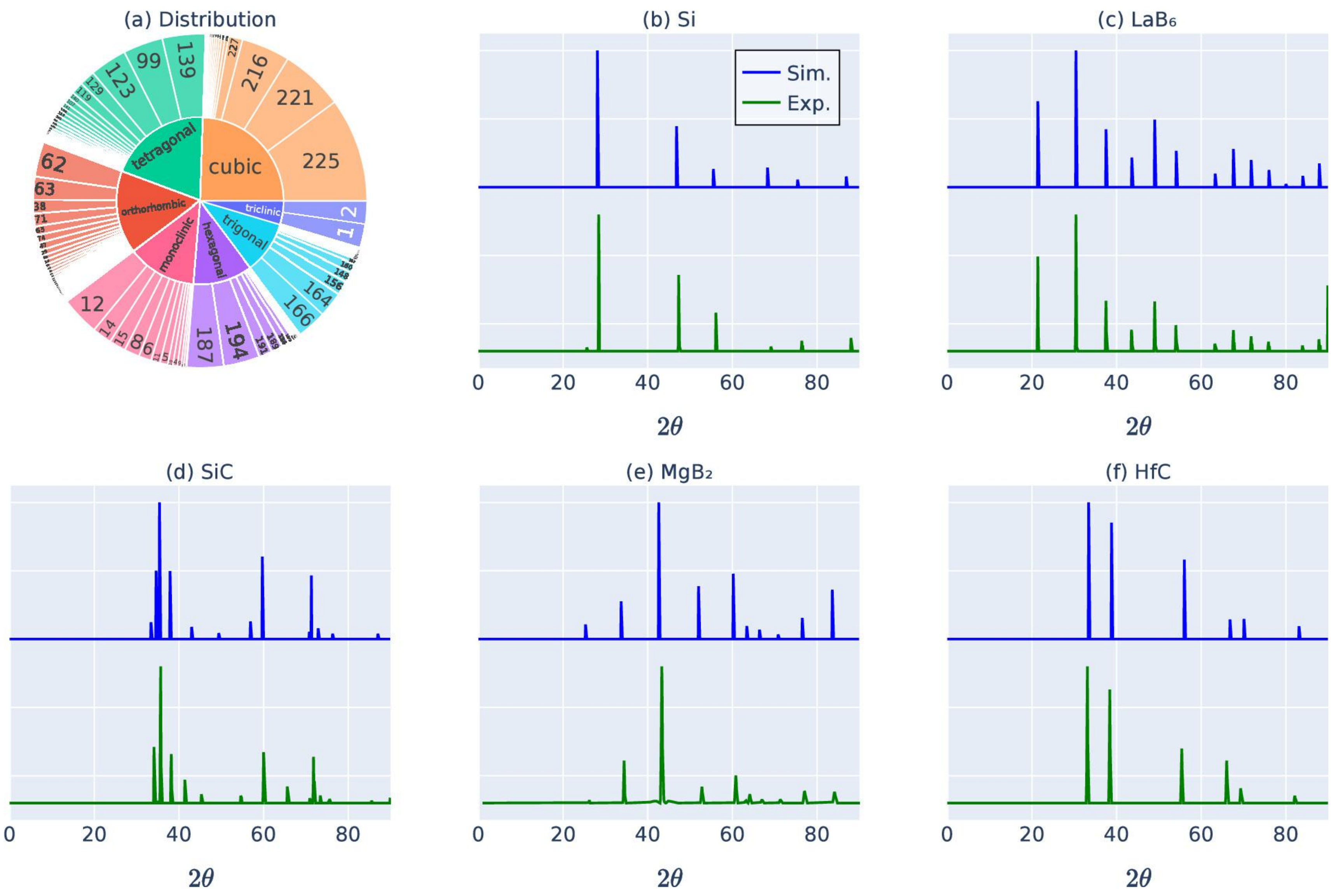
Example 6: Superconductor App





https://atomgpt.org/supercon

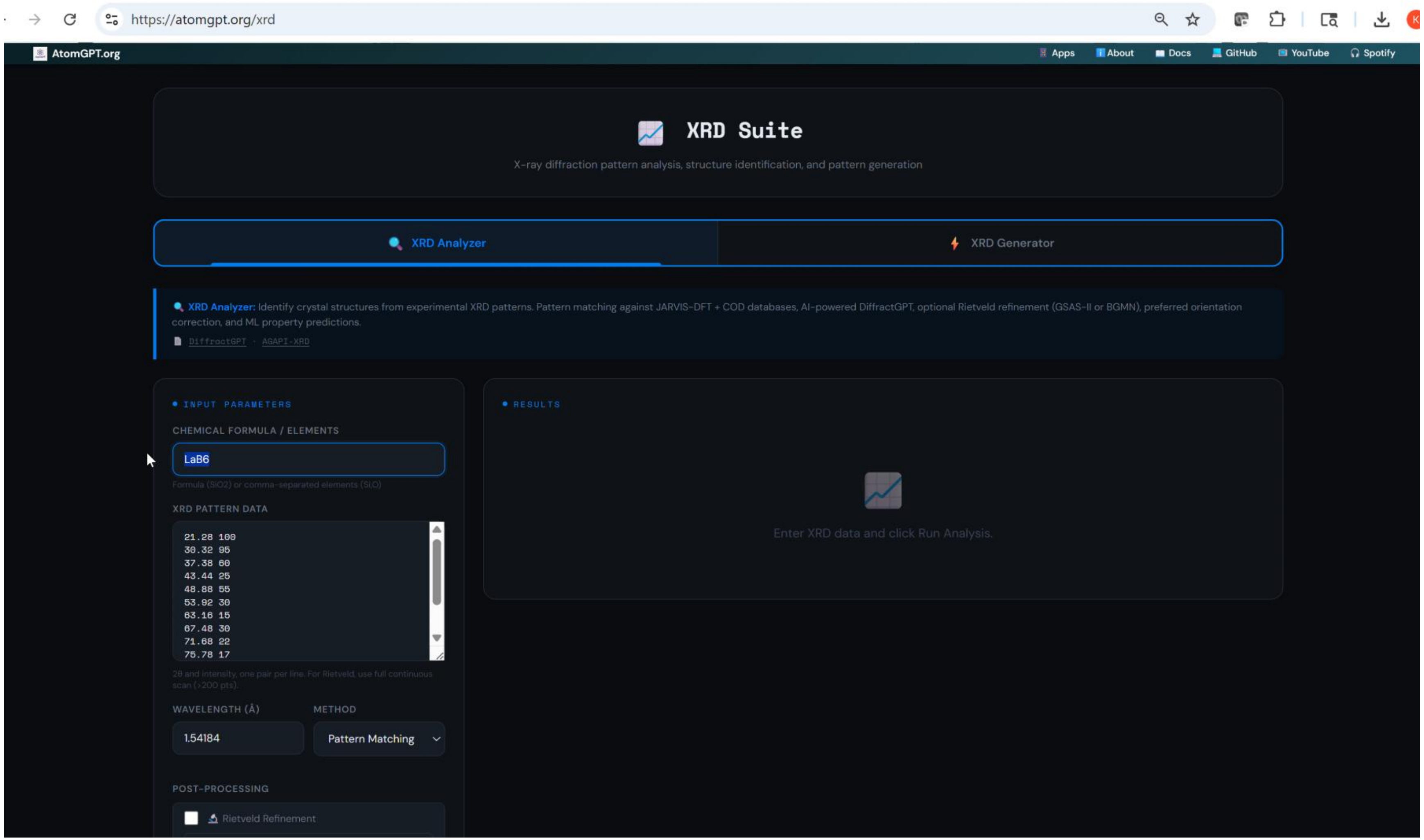
PXRD Database for Training

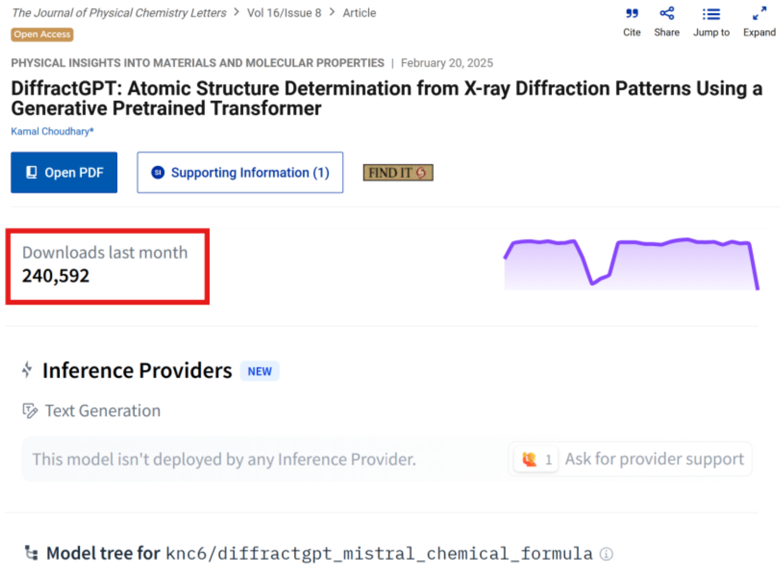



```python
class XRD(object):
    """Constryct an XRD class."""

    def __init__(
        self,
        wavelength=1.54184,
        thetas=[0, 180],
        two_theta_array=[],
        dhkl_array=[],
        intensity_array=[],
        scaling_factor=100,
        two_theta_tol=1e-5,
        intensity_tol=0.5,
        max_index=5,
    ):
        """Initialize class with incident wavelength."""
        self.wavelength = wavelength
        self.min2theta = np.radians(thetas[0])
        self.max2theta = np.radians(thetas[1])
        self.thetas = thetas
        self.two_theta_array = two_theta_array
        self.dhkl_array = dhkl_array
        self.intensity_array = intensity_array
        self.two_theta_tol = two_theta_tol
        self.intensity_tol = intensity_tol
```

Example 7: AI for XRD, DiffractGPT WebApp





Think DiffractGPT as AlphaFold for MSE

Similar to mainstream DeepSeek v3 downloads

https://atomgpt.org/xrd

AtomVision

MACHINE LEARNING AND DEEP LEARNING | March 1, 2023

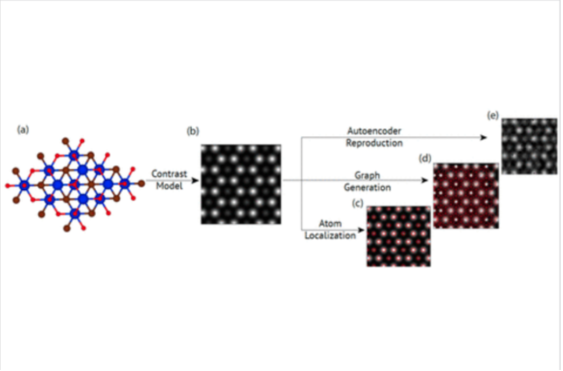
AtomVision: A Machine Vision Library for Atomistic Images

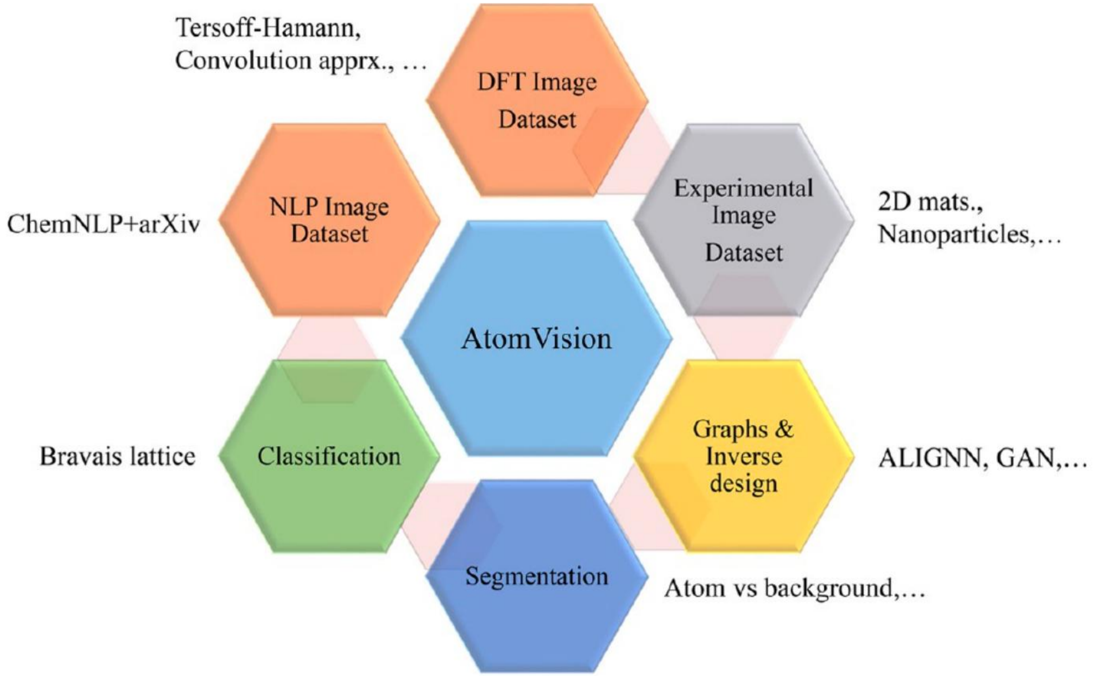
Kamal Choudhary*, Ramya Gurunathan, Brian DeCost, and Adam Biacchi

Other Access Options

Abstract

Computer vision techniques have immense potential for materials design applications. In this work, we introduce an integrated and general-purpose AtomVision library that can be used to generate and curate microscopy image (such as scanning tunneling microscopy and scanning transmission electron microscopy) data sets and apply a variety of machine learning techniques. To demonstrate the applicability of this library, we (1) establish an atomistic image data set of about 10 000 materials with large structural and chemical diversity, (2) develop and compare convolutional and atomistic line graph neural network models to classify the Bravais lattices, (3) demonstrate the application of fully convolutional neural networks using U-Net architecture to pixelwise classify atom versus background, (4) use a





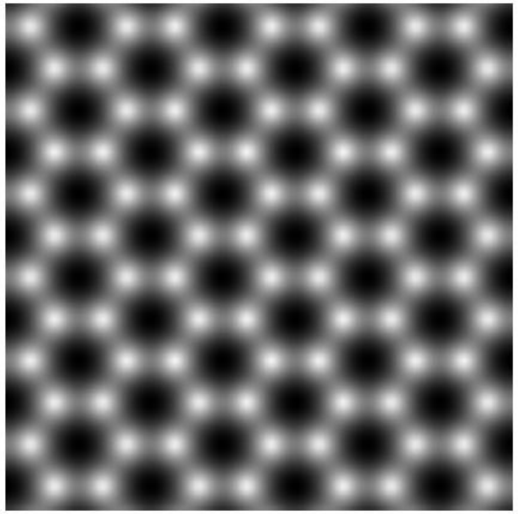
Various tasks with AtomVision

https://github.com/atomgptlab/atomvision

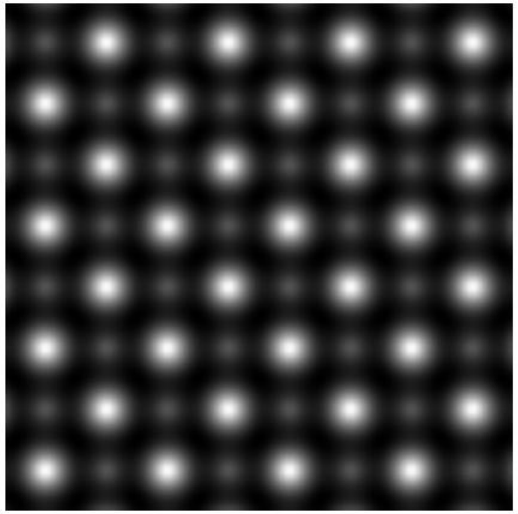
Scanning Transmission Electron Microscope Image

- Convolution approximation: accurate for **thin films** mainly (here 2D mats.)
- Based on Rutherford scattering model

$$I(\boldsymbol{r}) = R(\boldsymbol{r}, Z) \otimes \mathrm{PSF}(\boldsymbol{r}) \quad R(\boldsymbol{r}, Z) = \sum_{i=1}^{N} Z_i^{1.7} \delta(\boldsymbol{r} - \boldsymbol{r}_i)$$



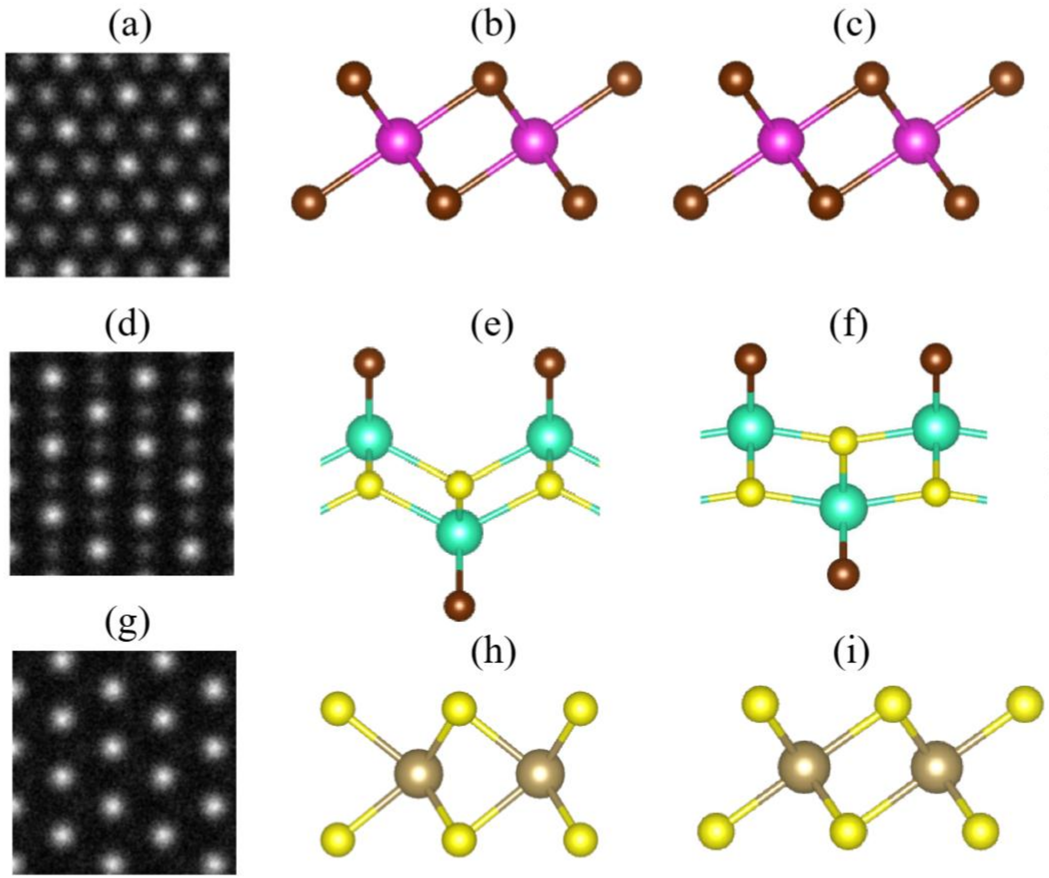
C: JVASP-667



FeTe: JVASP-6667

PPdSe: JVASP-6316

MicroscopyGPT: AtomGPT for Images



$CdBr_2$ ($P\bar{3}m1$),
JVASP-76512,
RMSE (As Gen.): 0.00
RMSE (ALIGNN-FF): 0.00

LuSBr (Pmmn),
JVASP-6205,
RMSE (As Gen.): 0.10,
RMSE (ALIGNN-FF): 0.05

TaS_2 ($P\bar{6}m2$),
JVASP-6070,
RMSE (As Gen.): 0.17
RMSE (ALIGNN-FF): 0.15

Prop.	Min.	Max.	KLD	EMD
a (Å)	2.38	14.398	0.06	1.67
b (Å)	2.51	15.64	0.07	0.91
γ ($\circ$)	60.0	120.0	0.02	0.54
Spg.	1	191	0.72	3.24
Weight (AMU)	24.82	2958.62	0.11	1.08

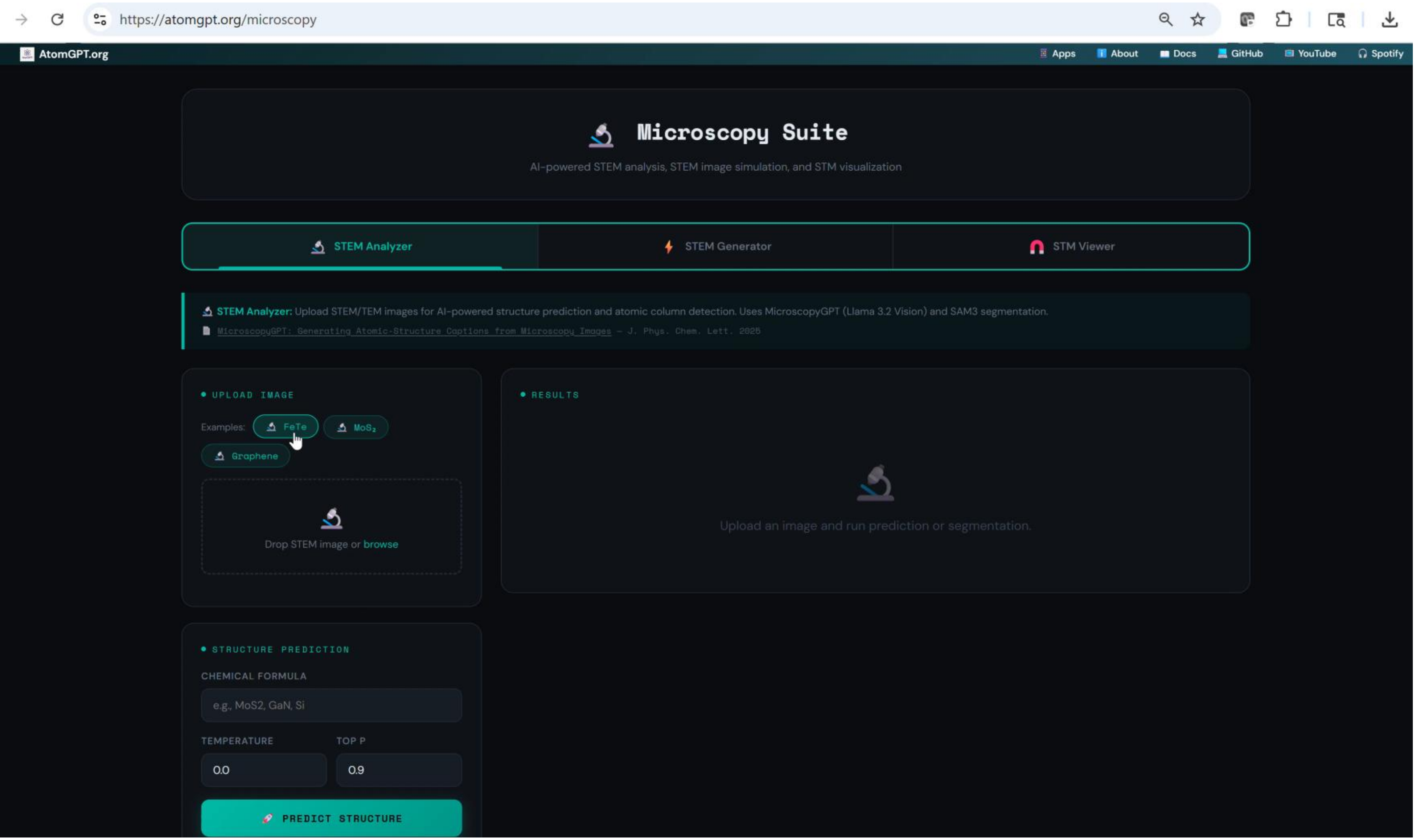
THE JOURNAL OF PHYSICAL CHEMISTRY LETTERS
A JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JPCL
Letter

MicroscopyGPT: Generating Atomic-Structure Captions from Microscopy Images of 2D Materials with Vision-Language Transformers

Kamal Choudhary*

Cite This: J. Phys. Chem. Lett. 2025, 16, 7028–7035

Example 8: MicroscopyGPT, Image Segmentation



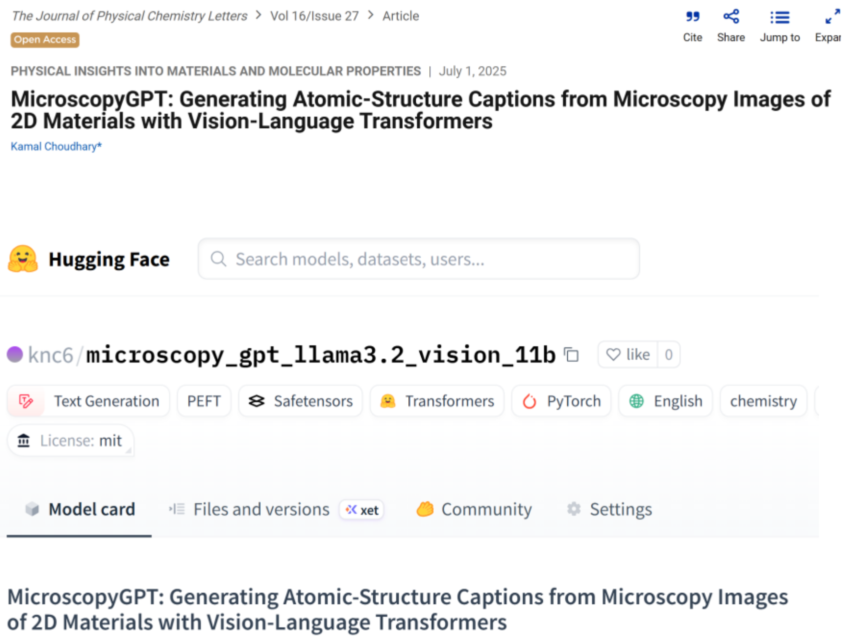
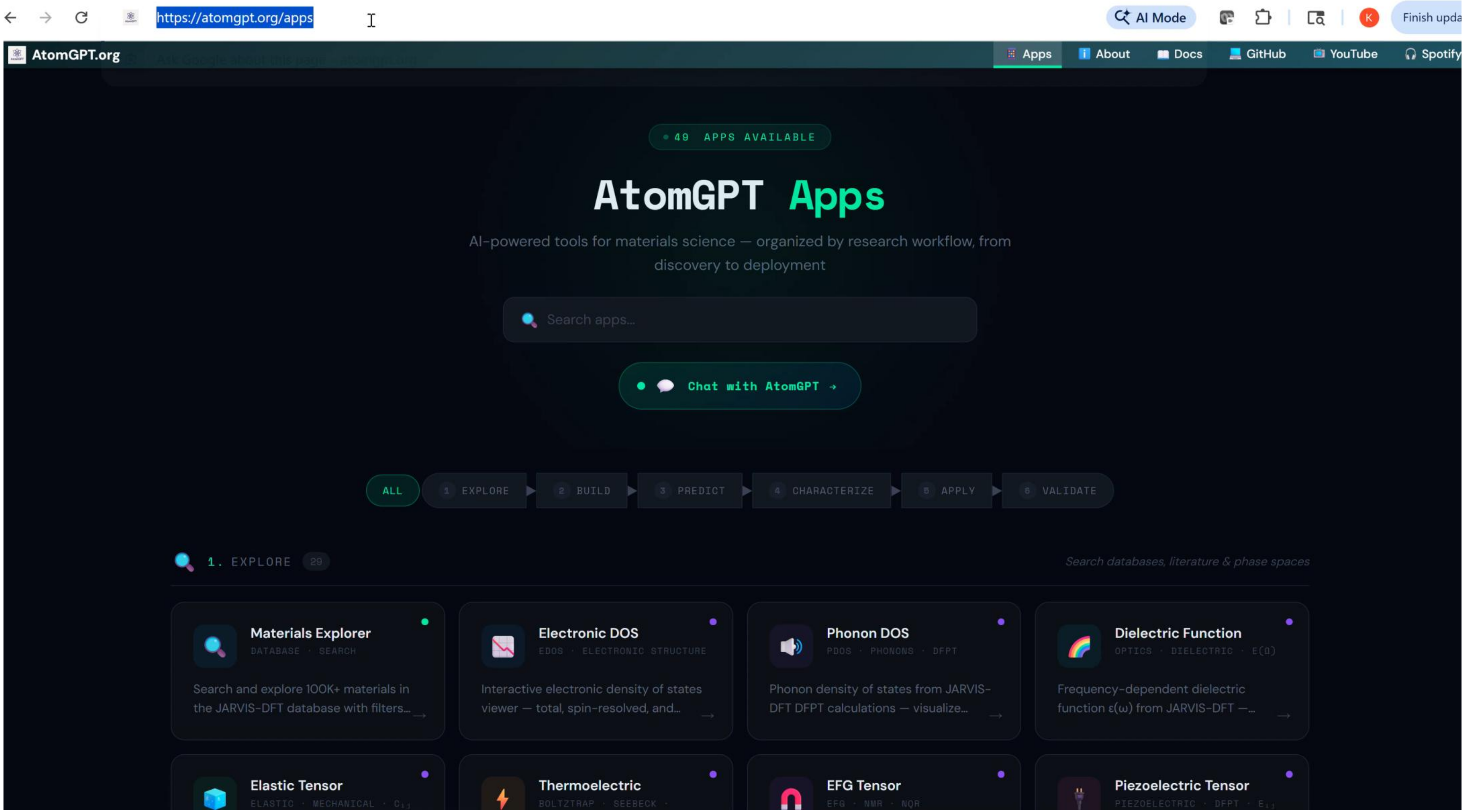


Image Segmentation with Meta-SAM3

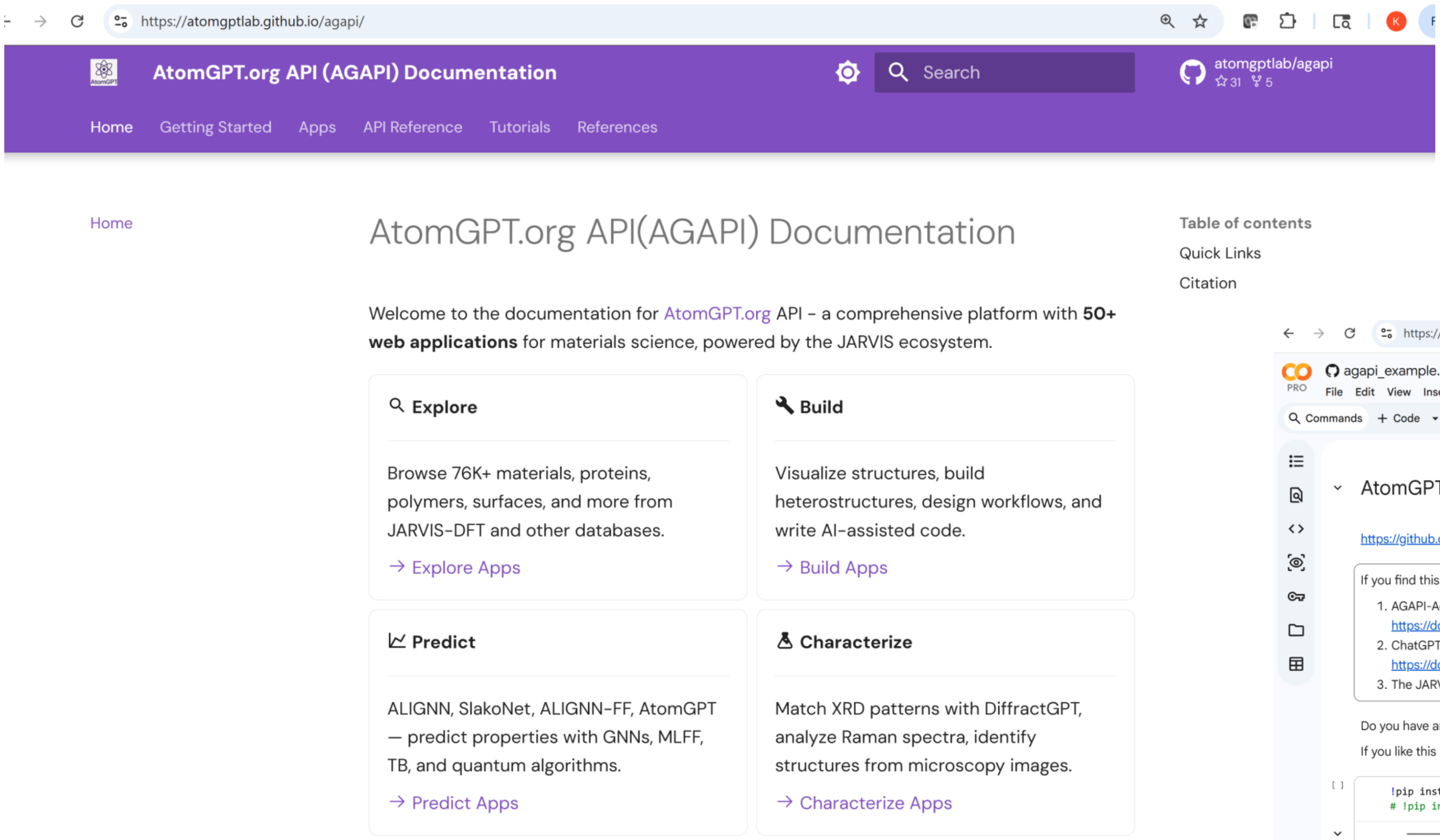
https://atomgpt.org/microscopy

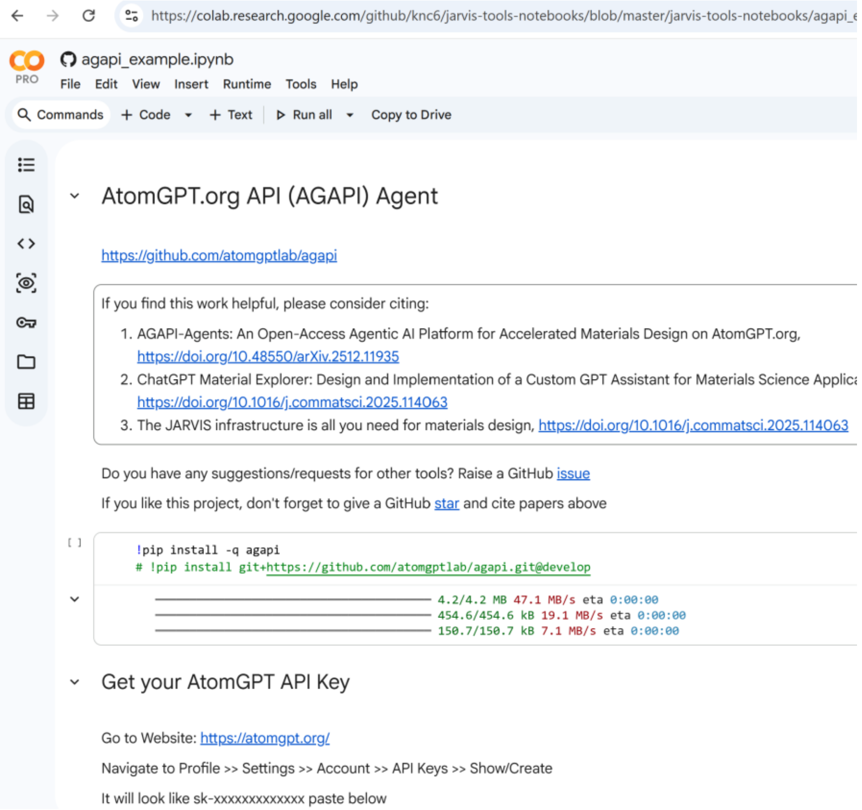
50+ Domain Specific APIs as WebApps



https://atomgpt.org/apps

API Tools Documentation





https://atomgptlab.github.io/agapi/

https://colab.research.google.com/github/knc6/jarvis-tools-notebooks/blob/master/jarvis-tools-notebooks/agapi_example.ipynb

Example Query for Database Search

```
ag5=agent.query_sync("How many materials have Tc_supercon data?",render_html=True)
```

Answer:

There are **1,058** materials in the JARVIS-DFT database that have a recorded superconducting transition temperature (`Tc_supercon`).

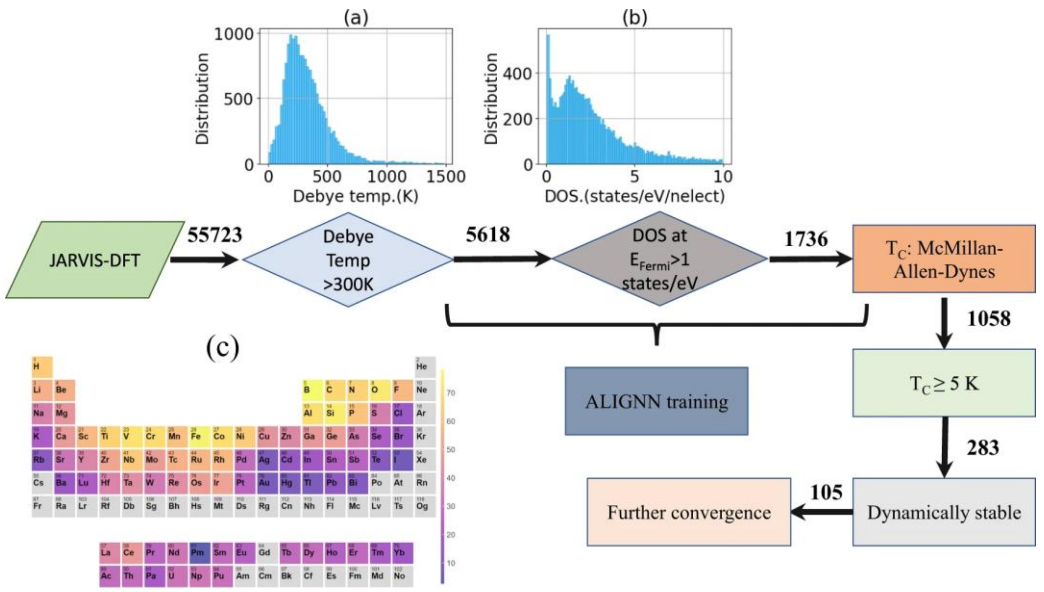
nature > npj computational materials > articles > article

Article | Open access | Published: 22 November 2022

Designing high-T_C superconductors with BCS-inspired screening, density functional theory, and deep-learning

Kamal Choudhary & Kevin Garrity

npj Computational Materials 8, Article number: 244 (2022) | Cite this article



If else statements vs. agentic search

Example Semiconductor Design: Multi-Agents/Prompts

```python
from agapi.agents import AGAPIAgent


ag25 = agent.query_sync("""
1. Find all GaN materials in the JARVIS-DFT database
2. Get the POSCAR for the most stable one (lowest formation energy)
3. Make a 2x1x1 supercell
4. Substitute one Ga with Al
5. Generate its powder XRD pattern using Cu K-alpha radiation
6. Report the top 10 strongest peaks
7. Optimize structure with ALIGNN-FF
8. Predict properties with ALIGNN
9. Calculate band structure with SlakoNet
10. Then create a summary table showing:
- Formation energy (ALIGNN prediction)
- Bandgap MBJ (ALIGNN prediction)
- Bandgap OptB88vdW (ALIGNN prediction)
- Bandgap (SlakoNet calculation)
- Bulk modulus (ALIGNN prediction)
- Shear modulus (ALIGNN prediction)
- XRD peaks

Display all values clearly.
""", auto_display_images=True, render_html=True, verbose=True, max_cont
```

The Journal of Physical Chemistry Letters > Vol 16/Issue 43 > Article

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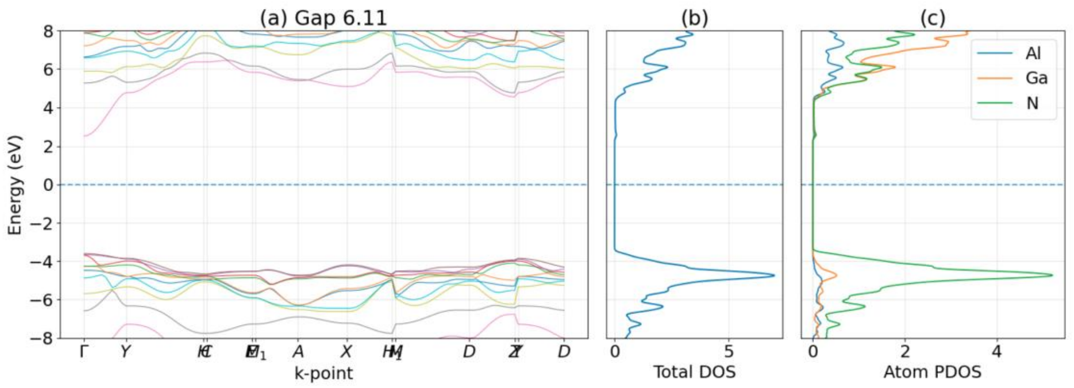
PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | October 17, 2025

SlaKoNet: A Unified Slater-Koster Tight-Binding Framework Using Neural Network Infrastructure for the Periodic Table

Kamal Choudhary*



✓ Displayed band structure image

Al-doped GaN ($AlGa_3N_4$) - 2 × 1 × 1 supercell (1 Al, 3 Ga, 4 N)

Example XRD Pattern

```
from agapi.agents import AGAPIAgent
ag27 = agent.query_sync("""
1. Find GaN in JARVIS-DFT
2. Get the most stable structure
3. Generate its powder XRD pattern using Cu K-alpha radiation
4. Report the top 10 strongest peaks
""", verbose=True,render_html=True)
```

```
[Iteration 1/15]
  Calling: query_by_elements
[Iteration 2/15]
  Calling: query_by_jid<|channel|>
[Iteration 3/15]
  Calling: generate_xrd_pattern
  [Truncated result from 36676 to 896 chars]
[Iteration 4/15]
**GaN – Most Stable Structure (JARVIS-DFT)**
- **JARVIS ID:** **JVASP-30**
- **Crystal System:** Hexagonal (space-group *P6₃mc*)
- **Lattice Parameters:**
  - *a* = 3.211 Å (derived from the POSCAR)
  - *c* = 5.240 Å
- **Formation Energy:** −0.571 eV / atom (thermodynamically the lowest-energy polymorph i
- **Bandgap (MBJ):** 3.08 eV

**Powder XRD Pattern (Cu Kα, λ = 1.54184 Å)**
The `generate_xrd_pattern` tool returned 8 distinct peaks within the 0–90° 2θ range.
Below are the **top 10 strongest peaks** (sorted by relative intensity). Since only 8 pea

| Rank | 2θ (°) | Intensity (relative) | d-spacing (Å) |
|------|--------|----------------------|---------------|
| 1 | 36.6 | 1.00 | 2.455 |
| 2 | 32.2 | 0.44 | 2.780 |
| 3 | 34.2 | 0.36 | 2.622 |
| 4 | 47.7 | 0.21 | 1.907 |
| 5 | 57.4 | 0.30 | 1.605 |
| 6 | 62.8 | 0.31 | 1.480 |
| 7 | 68.5 | 0.25 | 1.370 |
| 8 | 70.0 | 0.15 | 1.344 |
```

Outline

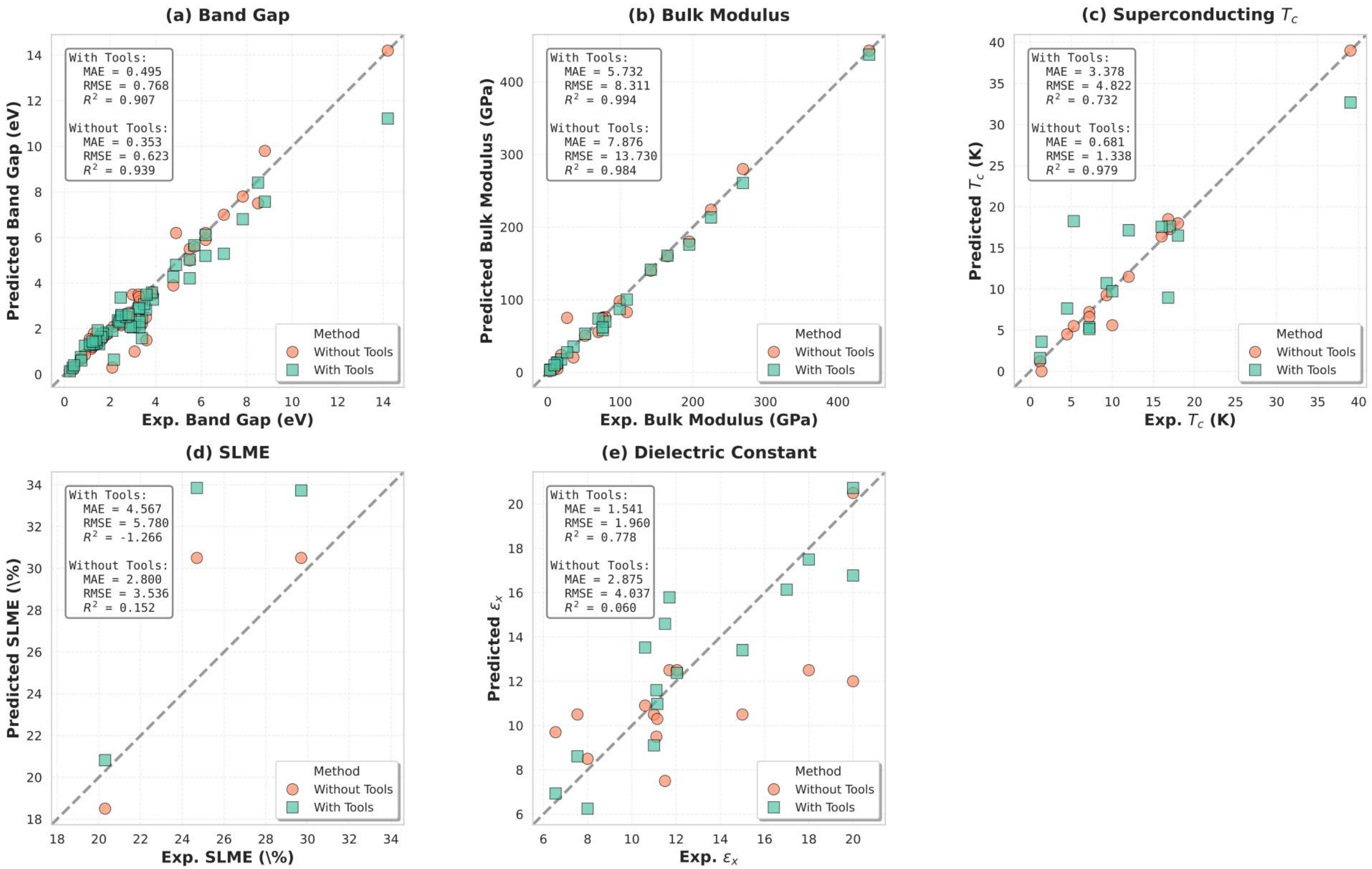
Overview of Agentic AI

ChatGPT Material Explorer

AtomGPT.org API Demos

<u>Benchmarking & Hackathon</u>

Agentic AI Benchmarking on Well-known Materials



Findings suggest that tool augmentation is not universally beneficial, its effectiveness depends
critically on three factors:

(1) database quality and coverage for the target property,

(2) standardization of computational protocols used to generate database entries, and

(3) whether the material and its properties are well-documented in existing literature.

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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | June 16, 2026

AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org

Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary*

Article | Open access | Published: 07 May 2024

JARVIS-Leaderboard: a large scale benchmark of materials design methods

Kamal Choudhary, Daniel Wines, Kangming Li, Kevin F. Garrity, Vishu Gupta, Aldo H. Romero, Jaron T. Krogel, Kayahan Saritas, Addis Fuhr, Panchapakesan Ganesh, Paul R. C. Kent, Keqiang Yan, Yuchao Lin, Shuiwang Ji, Ben Blaiszik, Patrick Reiser, Pascal Friederich, Ankit Agrawal, Pratyush Tiwary, Eric Beyerle, Peter Minch, Trevor David Rhone, Ichiro Takeuchi, Robert B. Wexler, Arun Mannodi-Kanakkithodi, Elif Ertekin, Avanish Mishra, Nithin Mathew, Mitchell Wood, Andrew Dale Rohskopf, Jason Hattrick-Simpers, Shih-Han Wang, Luke E. K. Achenie, Hongliang Xin, Maureen Williams, Adam J. Biacchi & Francesca Tavazza

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npj Computational Materials 10, Article number: 93 (2024) | Cite this article

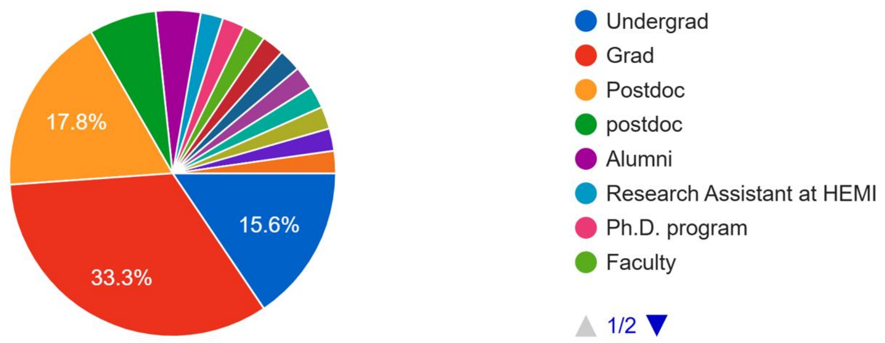
AAI4Science Hackathon

Agentic AI for Science (AAI4Science) Hackathon 2025

By **Data Science and AI Institute** Follow

Bloomberg Student Center · Baltimore, MD

Nov 6 from 1 pm to 3 pm EST



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MACHINE LEARNING
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Lessons learned from the 2025 Agentic AI for Science Hackathon

Jaehyung Lee, Harichandaan Neralla, Charles Campbell, Kent Zhang, Akshaya Ajith, Justin Ely, Hua Jiang, Zhijian Zhou, Nikhil Ramlukan, Xiaolong Li Show full author list

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.gitignore	Initial commit	3 m
LICENSE	Initial commit	3 m
README.md	Update README.md	3 m

README Apache-2.0 license

Agentic AI for Science (AAI4Science) Hackathon 2025

Date: Thursday, November 6, 2025
Location: Bloomberg Student Center, Room 204 (Hybrid)
Registration: Register here
Colab Notebook: Open in Colab
Submission Form: Google Form
AtomGPT.org API (AGAPI): GitHub Repository

Stay tuned for 2026 AAI4Science hackathon at JHU+virtual!
Signup at atomgpt.org

Let’s collaborate!

Laboratories

NIST:
F. Tavazza, B. DeCost, K. Garrity, C. Campbell, D. Wines, A. Biacchi, A. Hight Walker, S. Krylyuk, C. Camp Jr.

ORNL:
R. Vasudevan, D. Zhang, B. Sumpter, V. Sharma

ANL:
N. Anand

PNNL:
M. Ziatdinov

SLAC:
A. Mehta

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T. Frolov

Industry & Other

GE Aerospace:
G. Pilania

FDA:
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UTK/UT:
S. Kalinin, T. Liang, L. Vlcek

UF:
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Y. Cheng

Penn State:
S. Sinnott

Toronto:
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P. Acar

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ASU:
H. Zhuang

Texas A&M:
X. Qian

Missouri S&T:
A. Chernatynskiy

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A. Goyal

Central South:
J. Jiang, P. Wang

UNICAMP:
A. Fontes da Fonseca

U. Antwerpen:
M. Bercx, D. Lamoen

NCSU:
E. Bucholz

PSI:
M. Bercx

Our team

Jayhuyng Lee

C. Rhys Campbell

Justin Ely

Kent Zhang

Akshya Ajith

Shrijani

Buruganahalli

Frank Abel
(USNA)

https://choudhary.wse.jhu.edu/

Thank You!

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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | June 16, 2026

AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org

Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary*

Dr. Kamal Choudhary

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drkamal@jhu.edu